MAT327

2024-08-07

[1] James R. Munkres. 2000. *Topology* (2nd ed. ed.). Prentice Hall, Inc., Upper Saddle River, NJ.

1 Topology, Basis, Subbasis

Definition 1A. comparability

for topology $\mathcal{T}, \mathcal{T}'$ on set X
- $\mathcal{T} \subseteq \mathcal{T}'$ iff $\mathcal{T}$ is **coarser** iff $\mathcal{T}'$ is **finer**
- $\mathcal{T}$ and $\mathcal{T}'$ are **comparable** iff coarser or finer

1.1 Basis

Definition 11A. basis

for set X, system $\mathcal{B} \subseteq \mathcal{P}(X)$ is a **basis** of X iff all
- $\mathcal{B}$ covers X (i.e. X is open)
$$X \subseteq \bigcup \mathcal{B}$$
- $B_1 \cap B_2$ is open
$$\forall B_1, B_2 \in \mathcal{B} \quad \forall x \in B_1 \cap B_2 \quad \exists B_3 \in \mathcal{B} \quad x \in B_3 \subseteq B_1 \cap B_2$$

the topology **generated** by basis $\mathcal{B}$ is unions of basis elements
- elements of a basis are **basic open** sets

Theorem 11A. finer topology iff has subset basis

for basis $\mathcal{B}$ of $\mathcal{T}$ and basis $\mathcal{B}'$ of $\mathcal{T}'$

$$\mathcal{T} \text{ is coarser than } \mathcal{T}' \iff \forall x \in X \quad \forall B \in \eta(x) \quad \exists B' \in \mathcal{B}' \quad x \in B' \subseteq B$$

Proof.

($\Longrightarrow$) let any $x \in X$ and any basic $B \in \eta(x)$
- $\therefore B \in \mathcal{T} \subseteq \mathcal{T}'$
- $\therefore B$ is union of basis elements in $\mathcal{B}'$

($\Longleftarrow$) let any $U \in \mathcal{T}$, wts $U \in \mathcal{T}'$
- let any $x \in U$, thus $\exists B_x \in \mathcal{B} \quad x \in B_x \subseteq U$
- $\therefore \exists B'_x \in \mathcal{B}' \quad x \in B'_x \subseteq B$
- $\therefore U = \bigcup_{x \in U} B'_x$, so union of elements in $\mathcal{B}'$

$\square$

Lemma 11B. basis lemma

open collection $\mathcal{B} \subseteq \mathcal{P}(X)$ is basis iff

$$\forall x \in X \quad \forall U \in \eta(x) \quad \exists B \in \mathcal{B} \quad x \in B \subseteq U$$

1.2 Subbasis

Definition 12A. subbasis

system $\mathcal{S} \subseteq \mathcal{P}(X)$ is a **subbasis** of X iff covers X
- the topology **generated** by $\mathcal{S}$ is unions of finite intersections

1.3 Exercises

Example 13A. $\mathbb{R} \subset \mathbb{R}_l$

Proof. let any $x \in \mathbb{R}$, any $(a, b) \in \eta(x)$
- $\therefore x \in [x, b) \subseteq (a, b)$

however, $0 \in [0, 1)$ but $\nexists U \in \eta_{\mathbb{R}}(0) \quad U \subseteq [0, 1) \in \eta_l(0)$ □

Example 13B. $\mathbb{R} \subset \mathbb{R}_K$

Proof.
- $\mathbb{R} \subseteq \mathbb{R}_K$ since basis is subset
- $\mathbb{R} \subset \mathbb{R}_K$ since $\nexists U \in \eta_{\mathbb{R}}(0) \quad U \subseteq (-1, 1) - K \in \eta_K(0)$ □

Example 13C. (§13.6) $\mathbb{R}_l$ and $\mathbb{R}_K$ incomparable

Proof.
- $[2, 3)$ open in $\mathbb{R}_l$, but not in $\mathbb{R}_K$
- $\mathbb{R} - K$ open in $\mathbb{R}_K$, but not in $\mathbb{R}_l$ □

Example 13D. (§13.8) $\mathcal{B} = \{(p, q) : p, q \in \mathbb{Q}\}$ generates $\mathbb{R}$

Proof. $\forall x \in \mathbb{R} \quad \forall (a, b) \in \eta(x) \quad \exists (p, q) \subseteq (a, b) \quad (p, q) \in \eta(x)$ by density □

2 Order Topology, Finite Product

2.1 Order Topology

Example 21A. open rays (both direction) form subbasis of order topology in X

Proof.

wts (a, ∞) is open

- case X has maximum, then $(a, \infty) = (a, \max X]$ is open
- case X has no maximum, then $(a, \infty) = \bigcup_{b>a} (a, b)$ is open

wts open rays generates same topology $\mathcal{T}_R$ as order topology $\mathcal{T}_o$

1. $\mathcal{T}_R$ is coarser than $\mathcal{T}_o$, since open rays are open in order topology
2. $\mathcal{T}_0$ is coarser than $\mathcal{T}_R$, since
   - $[\min X, b)$ and $(a, \max X]$ are open rays
   - $(a, b) = (-\infty, b) \cap (a, \infty)$, so open in $\mathcal{T}_R$

□

2.2 Product Topology

Theorem 22A. basis of product topology

for spaces $(X, \mathcal{B}_X)$ and $(Y, \mathcal{B}_Y)$

$$\mathcal{B} = \{B_X \times B_Y : B_X \in \mathcal{B}_X \text{ and } B_Y \in \mathcal{B}_Y\}$$

is basis of $X \times Y$

Proof.

let any open $(x, y) \in X \times Y$ and any $U \in \eta(x, y)$

- $\therefore \exists U_X \in \eta(x) \quad \exists U_Y \in \eta(y) \quad (x, y) \in U_X \times U_Y$ (def of product topology)
- $\therefore \exists B_X \in \mathcal{B}_X \quad x \in B_X \subseteq U_X$ and $\exists B_Y \in \mathcal{B}_Y \quad y \in B_X \subseteq U_Y$
- $\therefore (x, y) \in B_X \times B_Y$

□

Theorem 22B. projection preimage is subbasis

for spaces $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$

$$\mathcal{S} = \{\pi_1^{-1}[U] : U \in \mathcal{T}_X\} \cup \{\pi_2^{-1}[V] : V \in \mathcal{T}_Y\}$$

is subbasis of $X \times Y$

Proof.

1. $\pi_1^{-1}[U] = U \times Y$ is open in product topology, similar for $\pi_2^{-1}[V]$
2. for any basic $U \times V$ in product topology

$$\begin{aligned} U \times V &= (U \cap X) \times (Y \cap V) \\ &= (U \times Y) \cap (X \times V) \\ &= \pi_1^{-1}[U] \cap \pi_2^{-1}[V] \end{aligned}$$

$\therefore U \times V$ open in $\mathcal{T}_S$ □

3 Subspace, Closed, Limit, Hausdorff

3.1 Subspace Topology

Definition 31A. subspace topology

for space $(X, \mathcal{T}_X)$ and $A \subseteq X$

$$\mathcal{T}_A \stackrel{\text{def}}{=} \{U \cap A : U \in \mathcal{T}_X\}$$

is the **subspace** topology of A

Theorem 31A. space basis to subspace basis

for space $(X, \mathcal{B}_X)$

$$\mathcal{B}_A = \{B \cap A : B \in \mathcal{B}_X\}$$

is a basis of $A \subseteq X$

Theorem 31B.
- $A \in \mathcal{T}_X \wedge U \in \mathcal{T}_A \implies U \in \mathcal{T}_X$
- A and B subspace X and Y, then $A \times B$ subspace $X \times Y$

3.2 Closed Set

Definition 32A. closed set

$$K \subseteq X \text{ closed} \iff X - K \text{ open}$$

Theorem 32A. close sets in space

for space X
- $\varnothing, X$ closed
- intersection of closed is closed
- finite union of closed is closed

Theorem 32B. for A subspace X
- K closed in $A \iff B$ is intersection of closed sets in X with A
- K closed in A and A closed in $X \implies K$ closed in X

3.3 Interior and Closure

Definition 33A. interior

for space X, the **interior** of $A \subseteq X$ is

$$\operatorname{Int}_X A \stackrel{\text{def}}{=} A^\circ \stackrel{\text{def}}{=} \bigcup \{U \subseteq X : U \subseteq A \text{ and } U \text{ open}\}$$

- i.e. union of open sets contained in A

$$a \in A^\circ \iff \exists U \in \eta(a) \quad U \subseteq A$$

Definition 33B. closure

for space X, the **closure** of $A \subseteq X$ is

$$\mathrm{cl}_X A \stackrel{\text{def}}{=} \overline{A} \stackrel{\text{def}}{=} \bigcap \{K \subseteq X : A \subseteq C \text{ and } C \text{ closed}\}$$

- i.e. intersection of closed sets containing A

$$a \in \overline{A} \iff \forall U \in \eta(a) \quad \exists a \in U \cap A$$

Theorem 33A. interior vs closure

for space X and $A \subseteq X$

$$A^\circ \subseteq A \subseteq \overline{A}$$

- $A^\circ = A \iff A$ is open
- $A = \overline{A} \iff A$ is closed

3.4 Boundary and Limit Points

Definition 34A. boundary

for space X, the boundary of $A \subseteq X$ is

$$\partial A \stackrel{\text{def}}{=} \overline{A} \cap \overline{A^c}$$

that is

$$a \in \partial A \iff \forall U \in \eta(a) \quad U \text{ intersects } A \text{ and } A^c$$

Theorem. $\partial A = \partial(X - A)$

Definition 34B. ~~accumulation~~ limit point

point $x \in X$ is a **limit point** of $A \subseteq X$ iff

$$\forall U \in \eta(x) \quad \exists a \in U^* \cap A \qquad \text{for } U^* = U - \{x\}$$

- A' is the set of limit points of A

Theorem 34A. closure is union with limit points

for $A \subseteq X$

$$\overline{A} = A \cup A'$$

Corollary. A closed $\iff A' \subseteq A$

3.5 Convergence and Hausdorff

Definition 35A. convergence

sequence $(x_n \in X)_{n\in\mathbb{N}}$ **converges** to $x \in X$ iff

$$\forall U \in \eta(x) \quad \exists N \in \mathbb{N} \quad \forall n \geqslant N \quad a_n \in U$$

or $\forall U \in \eta(x) \quad \{x_n\} \subseteq^* U$

Definition 35B. Hausdorff

space X is **Hausdorff** iff
- any distinct $a, b \in X$ has disjoint open neighborhoods

Theorem 35A.

for Hausdorff space X

Proof. singletons are closed

for any $a \in X$, we have

$\because \forall b \in X - \{a\} \quad \exists U_b \in \eta(b) \quad a \notin U_b$ thus $b \notin \overline{\{a\}}$
$\therefore \overline{\{a\}} = \{a\}$, thus closed □

Proof. every sequence converges to at most one point

let $A = \{a_n\}$, assume $a_n \to a$ and $a_n \to b$

BWOC assume $a \neq b$, let separating $U_a \in \eta(a)$ and $U_b \in \eta(b)$

$\therefore A \subseteq^* U_a$
$\therefore A \cap U_b \subseteq A - U_a$ where $A - U_a$ finite
$\therefore A \cap U_b$ finite, contradicts $A \subseteq^* U_b$ □

3.6 Exercises

Example 36A. §17.6

- $A \subseteq B \implies \overline{A} \subseteq \overline{B}$
- $\overline{A \cup B} = \overline{A} \cup \overline{B}$
- $\bigcup_\alpha \overline{A_\alpha} \subseteq \overline{\bigcup_\alpha A_\alpha}$
- $(A \cap B)^\circ = A^\circ \cap B^\circ$
- $A^\circ \cup B^\circ \subseteq (A \cup B)^\circ$

Example 36B. §17.13

$$\text{space } X \text{ Hausdorff} \iff \Delta \stackrel{\text{def}}{=} \{(x, x) : x \in X\} \text{ closed in } X^2$$

Proof. $(\implies)$ wts $\Delta' \subseteq \Delta$

let any $(a, b) \in \Delta'$, BWOC assume $a \neq b$
let $U_a \in \eta(a)$ and $U_b \in \eta(b)$ disjoint
$\therefore U_a \times U_b \in \eta(a, b)$, thus $\exists (x, x) \in (U_a \times U_b) \cap \Delta$
$\therefore x \in U_a \cap U_b$, contradicts disjoint □

Proof. ($\Longleftarrow$)

let any distinct $a, b \in X$
$\therefore (a, b) \in X^2 - D$ which is open
$\therefore \exists U \times V \in \eta(a, b) \quad U \times V \subseteq X^2 - D$ (basis of product topology)
BWOC assume $\exists x \in U \cap V$
$\therefore (x, x) \in U \times V$, contradicts $U \times V$ disjoints D □

Example 36C. §17.19

$$\overline{A} = A^\circ \sqcup \partial A$$

Proof.

wts disjoint, BWOC assume $\exists a \in A^\circ \cap \partial A$
- $\because A^\circ \in \eta(a)$ and $a \in \partial A \subseteq \overline{A^c}$
- $\therefore A^\circ$ intersects A^c, contradicts $A^\circ \subseteq A$ (def of closure)

($\Longrightarrow$) let any $a \in \overline{A} - \partial A$, wts $a \in A^\circ$
- $\therefore a \notin \partial A$ has some $U \in \eta(a)$ disjoints A or A^c
- $\because U$ intersects A $a \in \overline{A}$
- $\therefore U$ disjoints A^c, thus $a \in U \subseteq A$

($\Longleftarrow$)
1. $\partial A \subseteq \overline{A}$ by definition
2. $A^\circ \subseteq A \subseteq \overline{A}$ □

Example 36D. §17.19

$$\partial A = \varnothing \Longleftrightarrow A \text{ clopen}$$

Proof.

($\Longrightarrow$) assume $\partial A = \varnothing$

$$\therefore \quad \overline{A} = A^\circ \cup \partial A = A^\circ$$
$$\because A^\circ \subseteq A \subseteq \overline{A}$$
$$\therefore A^\circ = A = \overline{A}$$

($\Longleftarrow$) assume A clopen

$$\therefore \quad A = \overline{A} = A^\circ \sqcup \partial A = A \sqcup \partial A$$
$$\therefore \partial A = \varnothing$$

□

Example 36E. every sequence converges to at most one point $\not\Rightarrow$ Hausdorff

Proof. consider cocountable $\mathbb{R}$

BWOC assume U, V separates
$\therefore \mathbb{R} = (U \cap V)^c = U^c \cup V^c$, which is finite union of countable

let $a_n \to a$ and $b \neq a$, wts $a_n \not\to b$

$\because \mathbb{R} - \{b\} \in \eta(a)$
$\therefore \exists N \forall n \geqslant N \quad a_n \neq b$
let $A = \{a_N, a_{N+1}, \cdots\}$
$\therefore \mathbb{R} - A \in \eta(b)$, but $\mathbb{R} - A \not\subseteq^* \{a_n\}$ □

4 Continuous Map, Product Topology

Definition 4A. open map, closed map

for function $f : (X, \mathcal{T}_X) \to (Y, \mathcal{T}_Y)$
- f is **open** iff maps open sets to open sets
- f is **closed** iff maps closed sets to closed sets

Corollary. f open iff f^{-1} continuous

4.1 Continuous Map

Definition 41A. continuous map

function $f : X \to Y$ **continuous** TFAE
(1) preimage of open is open
(2) preimage of basis is open
(3) preimage of subbasis is open
(4) preimage of closed is closed
(5) $\forall x \in X \quad \forall V \in \eta(f(x)) \quad \exists U \in \eta(x) \quad f[U] \subseteq V$
(6) $\forall A \subseteq X \quad f\left[\overline{A}\right] \subseteq \overline{f[A]}$

Proof. continuous $\Longrightarrow$ (6)

let any $f(x) \in f\left[\overline{A}\right]$
let any $V \in \eta(f(x))$, wts V intersects $f[A]$
$\therefore f^{-1}[V] \in \eta(x)$ (f continuous)
$\therefore \exists a \in f^{-1}[V] \cap A$ ($x \in \overline{A}$)
$\therefore f(a) \in V \cap f[A]$ □

Proof. (6) $\Longrightarrow$ continuous

let any closed $K \subseteq Y$, wts $f^{-1}[K]$ closed
$\therefore f\left[\overline{f^{-1}[K]}\right] \subseteq \overline{(f \circ f^{-1})[K]} \subseteq \overline{K} = K$
$\therefore \overline{f^{-1}[K]} \subseteq f^{-1}[K]$ □

Example 41A. continuous maps

- constant map $f(x) = a$
- inclusion $i : A \subseteq X \hookrightarrow X$
- composition of continuous
- restriction of continuous
- projection

Theorem 41A. pasting lemma

for space $X = A \cup B$ where A, B both open/closed
- if $f : A \to Y$ and $g : B \to Y$ agree on common domain
- then $f \cup g : X \to Y$ continuous

Proof. $(f \cup g)^{-1}[U] = f^{-1}[U] \cup g^{-1}[U]$ □

Corollary. if $f \restriction U_\alpha$ continuous, then f continuous

Theorem 41B. locally continuous implies continuous

for $f : X = \bigcup_\alpha U_\alpha \to Y$

$$\forall \alpha (f \restriction U_\alpha \text{ continuous}) \implies f \text{ continuous}$$

Proof. $f^{-1}[V] = \bigcup_\alpha (f \restriction U_\alpha)^{-1}[V]$ □

4.2 Homeomorphism

Definition 42A. homeomorphism

- function $f : X \xrightarrow{\sim} Y$ is **homeomorphsim** iff f and f^{-1} continuous
- function $f : X \hookrightarrow Y$ is **imbedding** iff f and f^{-1} continuous

Theorem 42A. homeomorphism iff [open iff image open]

function $f : X \xrightarrow{\sim} Y$ is homeomorphism iff for any $A \subseteq X$

$$A \text{ open} \iff f[A] \text{ open}$$

Proof.

$(\implies)$
- if $A \subseteq X$ open, then $f[A] = (f^{-1})^{-1}[A]$ open
- if $f[A] \subseteq Y$ open, then $A = f^{-1}[f[A]]$ open

$(\impliedby)$
- if $B = f[f^{-1}[B]]$ open, then $f^{-1}[B]$ open, so f continuous

- if $A \subseteq X$ open, then $f[A] = (f^{-1})^{-1}[A]$ open, so f^{-1} continuous $\square$

4.3 Initial Topology

Definition 43A. initial topology

for set X and spaces $\{Y_\alpha\}$, the **initial** topology of $\{f_\alpha : X \to Y_\alpha\}$ is the coarest topology $\mathcal{T} \subseteq \mathcal{P}(X)$ in which each $f_\alpha : X \to Y_\alpha$ is continuous

Theorem 43A. inital topology generated by basis

the initial topology of $\{f_\alpha : X \to Y_\alpha\}$ is generated by subbasis

$$\mathcal{S} = \bigcup_\alpha \{f_\alpha^{-1}[V] : V \in \mathcal{T}_\alpha\}$$

Proof. let $\mathcal{T}_S$ be generated by $\mathcal{S}$, let $\mathcal{T}_F$ be initial topology of $\{f_\alpha : X \to Y_\alpha\}$

$\because$ $\mathcal{T}_S$ is topology on X where each f_α is continuous
$\therefore$ $\mathcal{T}_S$ is finer than $\mathcal{T}_\mathcal{F}$

let any topology $\mathcal{T}$ on X such that f_α continuous, wts $\mathcal{T}_S \subseteq \mathcal{T}$

$\because$ f_α continuous, thus $f_\alpha^{-1}[V] \in \mathcal{T}$ for any open $V \subseteq Y_\alpha$
$\therefore$ $\mathcal{S} \subseteq \mathcal{T}$, thus $\mathcal{T}_S \subseteq \mathcal{T}$ $\square$

4.4 Product and Box Topology

Definition 44A. cartesian product

the **cartesian product** of $\mathcal{X} = \{X_\alpha\}_{\alpha \in \Lambda}$ is

$$\begin{aligned}\prod_{\alpha \in \Lambda} X_\alpha &\overset{\text{def}}{=} \left\{(x_\alpha)_{\alpha \in \Lambda} : x_\alpha \in X_\alpha\right\} \\ &\overset{\text{def}}{=} \left\{f : \Lambda \to \bigcup \mathcal{X} \mid \forall \alpha \in \Lambda \quad f(\alpha) \in X_\alpha\right\}\end{aligned}$$

Definition 44B. product/box topology

the **product** topology on $X = \prod_\alpha X_\alpha$ is the initial topology of projections

$$\{\pi_\alpha : X \to X_\alpha \mid \alpha \in \Lambda\}$$

the **box** topology on $X_\square = \prod_\alpha X_\alpha$ is generated by basis

$$\left\{\prod_\alpha U_\alpha : U_\alpha \text{ open in } X_\alpha\right\}$$

Theorem 44A. basis of product/box topology

for topologies $\{(X_\alpha, \mathcal{B}_\alpha)\}_{\alpha\in\Lambda}$, let $X = \prod_\alpha X_\alpha$

- basis elements of X are in the form $\prod_\alpha U_\alpha$ where $\exists \Lambda_0 \subseteq [\Lambda]^{<\aleph_0}$ such that
$$\forall \alpha \in \Lambda_0 \quad U_\alpha \in \mathcal{B}_\alpha$$
$$\forall \alpha \notin \Lambda_0 \quad U_\alpha = X_\alpha$$
- basis elements of $X_\square$ are in the form $\prod_\alpha U_\alpha$ where each $U_\alpha \in \mathcal{B}_\alpha$

Theorem 44B. product of closure is closure of product

for $X = \prod_\alpha X_\alpha$ and $A_\alpha \subseteq X_\alpha$

- in box/product
$$\prod_\alpha \overline{A_\alpha} = \overline{\prod_\alpha A_\alpha}$$

Proof. $(\Longrightarrow)$

let any $x \in \prod_\alpha \overline{A_\alpha}$, any basic $U = \prod_\alpha U_\alpha \in \eta(x)$
$\because x_\alpha \in \overline{A_\alpha}$, choose $a_\alpha \in A_\alpha \cap U_\alpha$
$\therefore (a_\alpha)_{\alpha\in\Lambda} \in U \cap \prod_\alpha A_\alpha$ □

Proof. $(\Longleftarrow)$

let any $x \in \overline{A} = \overline{\prod_\alpha A_\alpha}$
let any $\alpha \in \Lambda$, wts $x_\alpha \in \overline{A_\alpha}$
let any $U_\alpha \in \eta(x_\alpha)$, thus $\pi_\alpha^{-1}[U_\alpha] \in \eta(x)$
$\therefore \exists a \in \pi_\alpha^{-1}[U_\alpha] \cap A$ $\qquad x \in \overline{A}$
$\therefore a_\alpha \in U_\alpha \cap A$, so $x_\alpha \in \overline{A_\alpha}$ □

Theorem 44C. product continuous iff each component continuous

for $f(y) = (f_\alpha(y))$
$$f : Y \to \prod_\alpha X_\alpha \text{ continuous} \Longleftrightarrow \forall \alpha \quad f_\alpha : Y \to X_\alpha \text{ continuous}$$

Proof. $(\Longrightarrow)$ in product/box: $f_\alpha = \pi_\alpha \circ f$ □

Proof. $(\Longleftarrow)$ in product:

let any $\alpha \in \Lambda$, any subbasis $\pi_\alpha^{-1}[U_\alpha]$ of $X = \prod_\alpha X_\alpha$ (for U_α open in X_α)
$\therefore f^{-1}[\pi_\alpha^{-1}[U_\alpha]] = (\pi_\alpha \circ f)^{-1}[U_\alpha] = f_\alpha^{-1}[U_\alpha]$ is open in Y □

Proof. $(\not\Longleftarrow)$ in box: consider $X = \mathbb{R}^\omega$

let $f : \mathbb{R} \to \mathbb{R}^\omega$ be $f(r) = (f_n(r))$ where $f_n(r) = 1$

- $\therefore f_n : \mathbb{R} \to \mathbb{R}$ continuous (constant functions continuous)

consider

$$V = \prod_{n \in \omega} (-1/n, 1/n) \in \eta_X(f(0))$$

BWOC assume $\exists \varepsilon > 0 \quad f[B(0, \varepsilon)] \subseteq V$

$\therefore$ for any $n \in \omega$

$$B(0, \varepsilon) = f_n[B(0, \varepsilon)] = (\pi_n^{-1} \circ f)[B(0, \varepsilon)] \subseteq \pi_n^{-1}[V] = (-1/n, 1/n)$$

a contradiction (notice n was arbitrary) $\square$

4.5 Exercises

Example 45A. §18.2

for continuous $f : X \hookrightarrow Y$ and $A \subseteq X$

$$x \in A' \implies f(x) \in f[A]'$$

Proof.

let any $V \in \eta(f(x))$
$\therefore f^{-1}[V] \in \eta(x)$ (continuous)
$\therefore \exists a \in f^{-1}[V]^* \cap A$ and $f(a) \neq f(x)$ (injective)
$\therefore f(a) \in V^* \cap f[A]$ $\square$

Example 45B. §18.8

for continuous $f, g : X \to (Y, \leqslant)$

$$h(x) = \min\{f(x), g(x)\}$$

is continuous
Proof.

let $A = \{x \in X : f(x) \leqslant g(x)\}$ and $B = \{x \in X : g(x) \leqslant f(x)\}$
wts A closed (similarly, B closed)

- let any $x \in X - A$
- case $\exists c \in (g(x), f(x))$
  - let $V_g = (-\infty, c) \in \eta(g(x))$ and $V_f = (c, \infty) \in \eta(f(x))$
  - let $U_f = f^{-1}[V_f], U_g = g^{-1}[V_g] \in \eta(x)$
  - let $U = U_f \cap U_g \in \eta(x)$
  - $\therefore \forall a \in U \quad g(a) < c < f(a)$
  - $\therefore x \in U \subseteq X - A$ as wanted
- case otherwise, thus $(g(x), \infty) = [f(x), \infty)$

- let $V_g = (-\inf, f(x)) \in \eta(g(x))$ and $V_f = [f(x), \infty) \in \eta(f(x))$
- $\therefore \forall a \in U \quad g(a) < f(x) \leqslant f(a)$
- $\therefore x \in U \subseteq X - A$ as wanted

wts $h : X \to Y$ continuous

$$\because h \restriction A = f \quad h \restriction B = g \quad X = A \cup B$$
$$\therefore h \text{ continuous}$$

□

Definition 45A. locally finite

system $\{A_\alpha \subseteq X\}$ is **locally finite** iff each $x \in X$ has a neighborhood that intersects at only finitely many A_αs

Example 45C. §18.9

let $f : X = \bigcup_\alpha A_\alpha \to Y$ where $\{A_\alpha\}$ is locally finite and A_α closed

$$\forall \alpha \quad f \restriction A_\alpha \text{ continuous} \implies f \text{ continuous}$$

Proof. wts $\forall x \in X \quad \forall V \in \eta_Y(f(x)) \quad \exists U \in \eta_X(x) \quad f[U] \subseteq V$

let any $x \in X$
$\therefore \exists U_0 \in \eta(x)$ that intersects only $\{A_i\}_{i=1}^n$
let subspace $A = \bigcup_i A_i$ and continuous $f_A = f \restriction A$ (paste)
$\therefore x \in U_0 \subseteq A$

let any $V \in \eta_Y(f(x))$
$\therefore \exists U_A = U_X \cap A \in \eta_A(x) \quad f[U_A] = f_A[U_A] \subseteq V$ $(f(x) \in \operatorname{ran} f_A)$
let $U = U_X \cap U_0 \in \eta_X(x)$

let any $f(a) \in f[U]$
$\therefore a \in U_X \cap U_0 \subseteq U_x \cap A = U_A$
$\therefore f(a) \in f[U_A] \subseteq V$ □

Example 45D. §18.13

let $A \subseteq X$, Hausdorff Y, continuous $f : A \to Y$

$$\exists! g : \overline{A} \to Y \quad f \subseteq g$$

Proof.

let any continunous $g_1, g_2 : \overline{A} \to Y$ be extensions of f
let any $x \in \overline{A}$
$\therefore \exists V_1 \in \eta(g_1(x)) \quad \exists V_2 \in \eta(g_2(x))$ separates

let $U = g_1^{-1}[V_1] \cap g_2^{-1}[V_2] \in \eta(x)$ (g_1, g_2 continuous)
$\therefore \exists a \in U \cap A$
$\therefore g_1(a) \in V_1$ and $g_2(a) \in V_2$, but $g_1(a) = f(a) = g_2(a)$, contradicts disjoint □

Example 45E. §19.6

let $X = \prod_{\alpha \in \Lambda} X_\alpha$ and sequence $(x_n \in X)$

$$x_n \to x \iff \forall \alpha \quad x_n(\alpha) \to x(\alpha)$$

Proof. ($\Longrightarrow$) product/box

let any α, any $U_\alpha \in \eta_\alpha(x(\alpha))$
$\therefore \pi_\alpha^{-1}[U_\alpha] \in \eta_X(x)$
$\therefore \exists N \forall n \geqslant N \quad x_n \in \pi_\alpha^{-1}[U_\alpha]$, thus $x_n(\alpha) \in (\pi_\alpha \circ \pi_\alpha^{-1})[U_\alpha] = U_\alpha$ □

Proof. ($\Longleftarrow$) product

let any $U = \prod_\alpha U_\alpha \in \eta_X(x)$ where $U_i = X_i$ for $i \notin \Lambda_0 \in [\Lambda]^{<\aleph_0}$
$\therefore \forall i \exists N_i \forall n > N \quad x_n(i) \in U_i$
let $N = \max_i \{N_i\}$
$\therefore \forall n > N \quad \forall \alpha \in \Lambda \quad x_n(\alpha) \in U_\alpha$, thus $x_n \in U$ □

Proof. ($\not\Leftarrow$) box

let $X = \mathbb{R}^\omega$ and $x_n = (\alpha/n)_{\alpha \in \omega}$
$\therefore \forall \alpha \in \omega \quad x_n(\alpha) = \alpha/n \to 0$ as $n \to \infty$
but $x_n \not\to 0$ because consider $U = (-1,1)^\omega \in \eta_X(0)$

$$\begin{aligned} \forall N \quad & \because x_N(N) = N/N = 1 \notin (-1,1) = U_N \\ & \therefore \quad x_N \notin U \end{aligned}$$

□

Example 45F. §19.7

let $\mathbb{R}^\infty \subseteq \mathbb{R}^\omega$ be set of eventually-zero sequences
Proof. $\overline{\mathbb{R}^\infty} = \mathbb{R}^\omega$ in product

let any $x \in \mathbb{R}^\omega$, any $U = \prod_n U_n \in \eta(x)$
$\therefore \exists N \forall n > N \quad U_n = \mathbb{R}$
$\therefore (x_1, \cdots, x_N, 0, 0, \cdots) \in U \cap \mathbb{R}^\infty$ □

Proof. $\overline{\mathbb{R}^\infty} = \mathbb{R}^\infty$ in box

let any $x \in \mathbb{R}^\omega - \mathbb{R}^\infty$
for any $n \in \omega$
- if $x_n = 0$, let $U_n = \mathbb{R} \in \eta(x_n)$
- if $x_n \neq 0$, let $U_n \in \eta(x_n)$ that separates 0 ($\mathbb{R}$ Hausdorff)

$\therefore x \in \prod_n U_n \subseteq \mathbb{R}^\omega - \mathbb{R}^\infty$ (there are infinitely many $n \in \omega$ such that $x_n \neq 0$) $\square$

5 Metric Topology

5.1 Metric and Metric Topology

Definition 51A. Metric Topology

A **metric** is $d : X^2 \to \mathbb{R}$ where

$$\begin{aligned} \forall x \forall y \quad & d(x,y) \geqslant 0 \text{ and } d(x,y) = 0 \iff x = y \\ \forall x \forall y \quad & d(x,y) = d(y,x) \\ \forall x \forall y \forall z \quad & d(x,z) \leqslant d(x,y) + d(y,z). \end{aligned}$$

The **ball** of radius $\varepsilon > 0$ centered at $x \in X$ is

$$B(x,\varepsilon) \stackrel{\text{def}}{=} \{y \in X : d(x,y) < \varepsilon\}.$$

The **metric space** (X,d) is generated by basis

$$\{B(x,\varepsilon) : x \in X \text{ and } \varepsilon > 0\}.$$

Space $(X,\mathcal{T})$ is **metrisable** iff there is a metric that induces $\mathcal{T}$.

Lemma 51A.

- Subspace of metrisable is metrisable — restriction map
- Metrisable $\implies$ Hausdorff — $B(x,\varepsilon/2)$ and $B(y,\varepsilon/2)$ for $\varepsilon = d(x,y)$
- $\prod_\alpha X_\alpha$ metrisable $\implies X_\alpha$ metrisable — X_α is homeomorphic subspace

Definition 51B. Standard Bounded Metric

For metric space (X,d), the **standard bounded metric** is $\overline{d} : X^2 \to \mathbb{R}$

$$\overline{d}(x,y) = \min\{d(x,y), 1\}$$

N.B. $\overline{d}$ induces the same topology as d

- $\{B(x,\varepsilon) : x \in X \quad \varepsilon < 1\}$ forms basis and are the same under d and $\overline{d}$

Definition 51C. Uniform Topology

For metric spaces $\{(X_\alpha, d_\alpha)\}$, the **uniform** topology on $X = \prod_\alpha X_\alpha$ is induced by

$$\overline{\rho}(x,y) = \sup_\alpha \overline{d}_\alpha(x_\alpha, y_\alpha)$$

Corollary. product coarser than uniform coarser than box

Theorem 51B. Continuity on Metrisable

Function $f : (X, d) \to (Y, d)$ continuous iff

$$\forall x \in X \quad \forall \varepsilon > 0 \quad \exists \delta > 0 \quad f[B(x, \delta)] \subseteq B(f(x), \varepsilon)$$

Theorem 51C. (X, d) metrisable $\implies X^\omega$ metrisable

$$D(x, y) = \sup_{i \in \omega} \frac{\overline{d}(x_i, y_i)}{i}$$

Proposition 51A. Product is finer than metric.

Proof. Let any basic $B_D(x, \varepsilon)$ open in metric, we show $\exists V \subseteq B(x, \varepsilon)$ open in product.
Let $N \in \mathbb{N}$ such that $1/N < \varepsilon$.
Let V open in product defined as

$$V = B_d(x_1, \varepsilon) \times \cdots \times B_d(x_N, \varepsilon) \times \mathbb{R} \times \mathbb{R} \times \cdots$$

We show $V \subseteq B(x, \varepsilon)$, for any $y \in V$, notice $\overline{d}(x_i, y_i) \leqslant 1$

$$\because \forall i \geqslant N \quad \frac{\overline{d}(x_i, y_i)}{i} \leqslant \frac{1}{N}$$

$$\therefore D(x, y) \leqslant \max\left\{\overline{d}(x_1, y_1)/1, \cdots, \overline{d}(x_N, y_N)/N, 1/N\right\} < \varepsilon$$

□

Proposition 51B. Metric is finer than product.

Proof. Let any basic $U = \prod_{n \in \omega} U_n$ with $U_n = \mathbb{R}$ for any $n \notin \Lambda_0 \in [\omega]^{<\omega}$.
Let any $x \in U$, we show $\exists B(x, \varepsilon) \subseteq U$.
Choose $B(x_i, \varepsilon_i) \subseteq U_i$ for any $i \in \Lambda_0$ where $\varepsilon_i < 1$. Define

$$\varepsilon = \min\{\varepsilon_i / i\}$$

We show $B(x, \varepsilon) \subseteq U$, let any $y \in B(x, \varepsilon)$,

$$\therefore \forall i \in \Lambda_0 \quad \overline{d}(x_i, y_i) \leqslant iD(x, y) < i\varepsilon \leqslant \varepsilon_i < 1$$

Therefore $d(x_i, y_i) = \overline{d}(x_i, y_i) < \varepsilon_i$, so $y \in U$. □

5.2 Countable Basis and First Countable Space

Definition 52A. Countable basis, First countable

Space X has **countable basis** at $x \in X$ iff

$$\exists\{U_n \in \eta(x)\}_{n\in\omega} \quad \forall U \in \eta(x) \quad \exists n \quad U_n \subseteq U.$$

Space X is **first countable** iff every point has a countable basis.

Corollary. Metrisable spaces are first countable. let $U_n = B(x, 1/n)$

5.3 Sequence Lemma and Sequential Space

Lemma 53A. Sequence Lemma

For any space X, any $A \subseteq X$, any $x \in X$

$$\exists(a_n) \subseteq A \quad a_n \to x \Longrightarrow x \in \overline{A}$$

Proof. Converse holds for first-countable spaces.

Let any $x \in \overline{A}$ with countable basis $\{U_n \in \eta(x)\}$.
Let $B_n = U_1 \cap \cdots \cap U_n \in \eta(x)$.
$\therefore$ $\{B_n\}$ is countable basis at x, and $B_{n+1} \subseteq B_n$
$\therefore$ for any $n \in \mathbb{N}$, choose $a_n \in B_n \cap A$ $(x \in \overline{A})$
$\therefore$ $\forall U \in \eta(x) \quad \exists N \quad B_N \subseteq U$, thus

$$\forall n \geqslant N \quad a_n \in B_n \subseteq B_N \subseteq U$$

□

Definition 53A. Sequential Space

Space X is **sequential** iff for any $A \subseteq X$

$$x \in \overline{A} \Longrightarrow \exists(a_n) \subseteq A \quad a_n \to x$$

- i.e. converse of sequence lemma holds

Corollary. First countable spaces are sequential.

Theorem 53B. Continuous Map Perserves Convergence

For function $f : X \to Y$

$$f \text{ continuous} \Longrightarrow (x_n \to x \Longrightarrow f(x_n) \to f(x))$$

Proof.

Let any $x_n \to x$. We show $f(x_n) \to f(x)$.
Let any $V \in \eta(f(x))$, thus $f^{-1}[V] \in \eta(x)$.
$\therefore$ $\exists N \forall n > N \quad x_n \in f^{-1}[V]$, thus $f(x_n) \in V$ □

Corollary 53C. Converse holds for sequential spaces.
Proof.

We show f continuous by $f[\overline{A}] \subseteq \overline{f[A]}$.

Let any $f(a) \in f[\overline{A}]$, any $V \in \eta(f(a))$.
$\therefore \exists (a_n) \subseteq A \quad a_n \to a.$ $(a \in \overline{A})$
$\therefore f(a_n) \to f(a)$, thus $\exists N \quad f(a_N) \in V \cap f[A]$. □

Example 53A. Nonsequential space, converse fails.

Let $\mathrm{id}_{\mathbb{R}} : \mathbb{R}_{\text{cocountable}} \to \mathbb{R}$
Proof.

We show in $\mathbb{R}_{\text{coco}}$, sequence $x_n \to x$ iff x_n is eventually x.
Assume x_n is not eventually x.
$\therefore \mathbb{R} - \{x_n : x_n \neq x\} \in \eta(x)$ but does not almost contain $\{x_n\}$. Thus $x_n \nrightarrow x$.
Thus $x_n \to x \implies f(x_n) \to f(x)$, but f is not continuous becaue $f^{-1}[(0,1)]$ is not open □

5.4 Pointwise, Uniform, Strong Convergence

Definition 54A. Space X and $\{f_n\} \subseteq [0,1]^X$

- The sequence (f_n) converges **pointwise** to $f : X \to [0,1]$ iff
$$\forall x \in X \quad \forall \varepsilon > 0 \quad \exists N \in \mathbb{N} \quad \forall n \geqslant N \quad f_n(x) \in B(f(x), \varepsilon)$$
- The sequence (f_n) converges **uniformly** to $f : X \to [0,1]$ iff
$$\forall \varepsilon > 0 \quad \exists N \in \mathbb{N} \quad \forall n \geqslant N \quad \forall x \in X \quad f_n(x) \in B(f(x), \varepsilon)$$
- The sequence (f_n) converges **strongly** to $f : X \to [0,1]$ iff
$$\forall r : X \to (0,1) \quad \exists N \in \mathbb{N} \quad \forall n \geqslant N \quad \forall x \in X \quad f_n(x) \in B(f(x), r(x))$$

Theorem 54A. Sequence $f_n \to f$ pointwise iff $f_n \to f$ in product topology.

Proof. $(\implies)$

Let any $U = \prod_x U_x \in \eta(f)$ where $U_x = Y$ for any $x \notin \Lambda_0 \in [X]^{<\omega}$.
For any $x \in \Lambda_0$, choose $B(f(x), \varepsilon_x) \subseteq U_x \in \eta(f(x))$.
Let $\varepsilon = \min_{x \in \Lambda_0} \varepsilon_x$. Thus $\forall x \in \Lambda_0 \quad \exists N_x \forall n \geqslant N_x \quad f_n(x) \in B(f(x), \varepsilon) \subseteq U_x$.
Let $N = \max_{x \in \Lambda_0} N_x$. Thus $\forall n \geqslant N \quad \forall x \in X \quad f_n(x) \in U_x$, thus $f_n \in U$. □

Proof. $(\impliedby)$

Let any $x \in X, \varepsilon > 0$.
Let $U = \prod_y U_y \in \eta(f)$ where $U_x = B(f(x), \varepsilon)$, otherwise $U_y = \mathbb{R}$.
$\therefore \exists N \forall n \geqslant N \quad f_n \in U$, thus $f_n(x) \in U_x = B(f(x), \varepsilon))$ □

Theorem 54B. Sequence $f_n \to f$ uniformly iff $f_n \to f$ in uniform topology.

Proof.

$(\implies) \forall \varepsilon > 0 \quad \exists N \quad \forall n \geqslant N \quad \forall x \in X \quad \overline{d}(f_n(x), f(x)) < \varepsilon/2$, that is,

$$\overline{\rho}(f_n, f) = \sup_{x \in X} \overline{d}(f_n(x), f(x)) \leqslant \varepsilon/2 < \varepsilon$$

$(\Longleftarrow) \forall \varepsilon > 0 \quad \exists N \quad \forall n \geqslant N \quad \overline{\rho}(f_n, f) < \varepsilon$, that is,

$$\forall x \in X \quad \overline{d}(f_n(x), f(x)) < \varepsilon$$

□

Theorem 54C. Sequence $f_n \to f$ strongly iff $f_n \to f$ in box topology.

Proof.

$(\Longrightarrow) \forall U = \prod_x B(f(x), \varepsilon_x) \in \eta(f)$, WLOG assume $\varepsilon_x < 1$, let $r(x) = \varepsilon_x$.
$\therefore \exists N \forall n \geqslant N \quad \forall x \in X \quad f_n(x) \in B(f(x), r(x))$, that is, $f_n \in U$.

$(\Longleftarrow) \forall r : X \to (0, 1)$, let $U = \prod_x B(f(x), r(x)) \in \eta(f)$.
$\therefore \exists N \forall n \geqslant N \quad f_n \in U$, that is, $\forall x \in X \quad f_n(x) \in B(f(x), r(x))$. □

Lemma 54D. $C(X; [0, 1]) \stackrel{\text{def}}{=} \left\{ f \in [0, 1]^X : f \text{ continuous} \right\}$ is closed subspace of $[0, 1]^X$ (uniform).
Proof.

Let any $f \in C'$, we show $f \in C$.
$\because [0, 1]^X$ uniform, thus metrisable, thus sequential.
$\therefore \exists (f_n \in C) \quad f_n \to f$.
Let any $x \in X$, any $\varepsilon > 0$, we show $\exists \delta > 0 \quad f[B(x, \delta)] \subseteq B(f(x), \varepsilon)$
$\because \exists N \quad \sup_y d(f_N(y), f(y)) < \varepsilon/3$. (uniform convergence)
$\therefore \exists \delta > 0 \quad f_N[B(x, \delta)] \subseteq B(f_N(x), \varepsilon/3)$. ($f_N$ continuous)
$\therefore$ for any $y \in B(x, \delta)$

$$\begin{aligned} d(f(x), f(y)) &\leqslant d(f(x), f_N(x)) + d(f_N(x), f_N(y)) + d(f_N(y), f(y)) \\ &\leqslant \varepsilon/3 + \varepsilon/3 + \varepsilon/3 = \varepsilon \end{aligned}$$

□

Example 54A. Lemma 54D fails on product topology

Let $f : [0, 1] \to [0, 1]$ where $f(1) = 1$ and $f(x) = 0$ for $x \neq 1$.
$\therefore f_n(x) = x^n$ converges to f pointwise, thus $f \in \overline{C}$. (sequence lemma)
But f is not continuous

5.5 Exercises

Example 55A. $X = \prod_\alpha X_\alpha$ is NOT metrisable for $|\Lambda| \geqslant \aleph_0$

BWOC assume X metrisable, thus first countable
let some $x \in X$ with countable basis $\{U_n \in \eta_X(x)\}$
$\therefore \forall n \quad U_n = \prod_\alpha U_n(\alpha)$ where $\exists \Lambda_n \in [\Lambda]^{<\aleph_0}$ such that

$$\forall \alpha \notin \Lambda_n \quad U_n(\alpha) = X$$

let $\Lambda' = \bigcup_n \Lambda_n$ be countable
$\therefore \exists \beta \in \Lambda - \Lambda'$, thus $U_n(\beta) = X$ for any n
let $U = \prod_\alpha V_\alpha \in \eta_X(x)$ where

$$V_\beta \neq X \in \eta(x_\beta)$$
$$V_\alpha = X \in \eta(x_\alpha) \quad \forall \alpha \neq \beta$$

$\therefore \nexists n \quad U_n \subseteq U$, namly $U_n(\beta) = X \nsubseteq V_\beta$

Example 55B. X not discrete $\implies X_\square^\omega$ not metrisable

Proof.

let $x_0 \in X$ such that $\{x_0\}$ not open
$\therefore x_0 \in \overline{X - \{x_0\}}$
let $x = (x_0, x_0, \cdots) \in X^\omega$
let $A = \{a \in X^\omega : \forall n \quad a_n \neq x_0\}$

BWOC assme $X_\square^\omega$ metrisable, thus sequential and Hausdorff
wts $x \in \overline{A}$

- let any $U = \prod_n U_n \in \eta(x)$
- $\therefore \forall n \exists a_n \neq x_0 \in U_n \in \eta(x_0)$
- $\therefore a \in A \cap U$

let any $(a_n \in A)$, wts $a_n \nrightarrow x$

- for any $n \in \omega$
  - $\because a_n(n) \neq x_0$ $\qquad (a_n \in A)$
  - $\therefore \exists U_n \in \eta(x_0) \quad a_n(n) \notin U_n$ $\qquad$ (Hausdorff)

let $U = \prod_n U_n \in \eta(x)$
$\therefore \forall n \quad a_n \notin U$, contradicts $X_\square^\omega$ is sequential $\square$

Example 55C. §20.3

let $(X, \mathcal{T}_d)$ generated by metric $d : X^2 \to \mathbb{R}$

$$\mathcal{T}_d \text{ is the coarest topology in which } d : X^2 \to \mathbb{R} \text{ is continuous}$$

Proof. $d : X^2 \to \mathbb{R}$ is continuous

let any $(x, y) \in X^2$, let $\delta = d(x, y)$
let any basic $(a, b) \in \eta(\delta)$
let $\varepsilon = \min\{\delta - a, b - \delta\}/2$
let $U = U_1 \times U_2 = B(x, \varepsilon) \times B(y, \varepsilon) \in \eta(x, y)$
wts $d[U] \subseteq (a, b)$, let any $\delta' = d(x', y') \in d[U]$

$$\begin{aligned} \therefore \delta' = d(x', y') &\leqslant d(x', x) + d(x, y) + d(y, y') \\ &< \varepsilon + \delta + \varepsilon = \delta + 2\varepsilon \leqslant \delta + b - \delta = b \\ \delta = d(x, y) &\leqslant d(x, x') + d(x', y') + d(y', y) \\ &< \varepsilon + \delta' + \varepsilon = \delta' + 2\varepsilon \leqslant \delta' + \delta - a \\ \therefore \qquad \delta' &> a \end{aligned}$$

thus $\delta' \in (a, b)$ □

Proof.

let any $\mathcal{T}$ in which $d : X^2 \to \mathbb{R}$ continuous, wts $\mathcal{T}_d \subseteq \mathcal{T}$

let any $x \in X$, any basic $B(x, \varepsilon) \in \eta_d(x)$
$\because d_x = d(x, -) : X \to R$ is continuous in $\mathcal{T}$ (projection continuous)
$(-\infty, \varepsilon)$ open in $\mathbb{R}$
$\therefore d_x^{-1}[(-\infty, \varepsilon)] = \{y \in X : d(x, y) < \varepsilon\} = B(x, \varepsilon)$ open in $\mathcal{T}$ □

Example 55D. §20.11

Given metric space (X, d)

Proposition 55A. $d'(x, y) = d(x, y)/(1 + d(x, y))$ is metric.

Proof.

First, define $f : \mathbb{R} \to \mathbb{R}$ by

$$f(x) = \frac{x}{1 + x} = 1 - \frac{1}{1 + x}$$

Notice $f : \mathbb{R} \to [0, 1)$ is
- nonnegative, bounded, increasing,
- $f(x) = 0 \iff x = 0$.

Thus $f \circ d = d'$ is
- nonnegative, symmetric, bounded,
- zero $\iff d$ is zero $\iff x = y$.

To show triangle inequality, first note that for $a, b \geqslant 0$,

$$f(a+b) - f(b) = \frac{a}{(1+b)(1+a+b)} \leqslant \frac{a}{1+a} = f(a)$$

$$\therefore \qquad f(a+b) \leqslant f(a) + f(b)$$

Lastly, because f is increasing and $f \circ d = d'$

$$f(d(x,z)) \leqslant f(d(x,y) + d(y,z)) \leqslant f(d(x,y)) + f(d(y,z))$$

Therefore d' is metric □

Proposition 55B. d' generates the same topology as d

Proof.

$\because$ f is d-continuous and $f^{-1}(y) = 1/(1-y)$ is d'-continuous.
$\therefore$ $d' = f \circ d$ is d-continuous, and $d = f^{-1} \circ d'$ is d'-continuous. □

Example 55E. §21.2

let $f : (X, d) \to (Y, d)$ such that

$$\forall x_1 \forall x_2 \quad d(x_1, x_2) = d(f(x_1), f(x_2))$$

then $f : X \hookrightarrow Y$ is an imbedding, called **isometric imbedding**

Proof.

wts f continuous

- let any $x_1 \in X$, any $\varepsilon > 0$

$$\begin{aligned} \therefore f[B(x_1, \varepsilon)] &= \{f(x_2) : d(x_1, x_2) < \varepsilon\} \\ &= \{f(x_2) : d(f(x_1), f(x_2)) < \varepsilon\} \\ &\subseteq \{y : d(f(x_1), y) < \varepsilon\} = B(f(x_1), \varepsilon) \end{aligned}$$

wts f injective

$$\begin{aligned} f(x_1) = f(x_2) &\implies d(f(x_1), f(x_2)) = 0 \\ &\implies d(x_1, x_2) = 0 \\ &\implies x_1 = x_2 \end{aligned}$$

□

6 Final Topology, Quotient Topology

6.1 Final Topology

Definition 61A. Final Topology

For topologies $\{X_\alpha\}$ and set Y, the **final topology** on Y induced by functions $\{f_\alpha : X_\alpha \to Y\}$ is

$$\{V \subseteq Y : \forall \alpha \quad f_\alpha^{-1}[V] \text{ is open in } X\}.$$

Corollary. The final topology is the finest topology in which each $f_\alpha : X_\alpha \to Y$ is continuous.

6.2 Quotient Map

Definition 62A. Saturated Set (Function)

For function $f : X \to Y$, subset $C \subseteq X$ is f-**saturated** iff (TFAE)
(1) $C = f^{-1}[f[C]]$
(2) $\exists S \subseteq Y \quad C = f^{-1}[S]$.

Definition 62B. Saturated Set (Relation)

For equivalence relation $E \subseteq X^2$, subset $A \subseteq X$ is E-**saturated** iff

$$A = \bigcup_{a \in A} [a]_E.$$

N.B. A is E-saturated iff C is preimage of its π fibres.

Definition 62C. Quotient Map

For spaces X and Y, a surjection $p : X \twoheadrightarrow Y$ is a **quotient map** iff (TFAE)
(1) V open/closed in $Y \iff p^{-1}[V]$ open/closed in X
(2) p maps saturated open/closed to open/closed
(3) p continuous and open/closed.

Corollary. Bijective quotient map is homeomorphism. $U = f^{-1}[f[U]]$

Example 62A. Not all quotient maps are open.

Let $X = [0,1] \cup [2,3]$ and $Y = [0,2]$ be subspaces of $\mathbb{R}$. The map $p : X \twoheadrightarrow Y$ by

$$p(x) = \begin{cases} x & x \in [0,1] \\ x - 1 & x \in [2,3] \end{cases}$$

is a continuous closed map, thus quotient map. But image of open $[0,1] \subseteq X$ is not open in Y.

Example 62B. Restriction of quotient map not quotient map of subspace.

Consider Example 62A and let $A = [0,1) \cup [2,3]$ be subspace of X. Notice $p \restriction A$ does not map the saturated open $[2,3]$ to an open set.

6.3 Quotient Topology

Definition 63A. Quotient Topology (Surjection)

For space X and Y, the **quotient topology** induced by surjection $p : X \twoheadrightarrow Y$ is the final topology on Y induced by p.

Corollary. The map p is a quotient map in the quotient topology.

Definition 63B. Quotient Space (Equivalence Relation)

For space X and quotient set X/E, the **quotient topology** on X/E is the quotient topology induced by the natural projection $\pi : X \to X/E$.

$$\pi(x) = [x]_E$$

P.S. MAT327 will use $\sim$ instead of E, use $X_\sim$ instead of $X/\sim$, use $p_\sim$ instead of π

Theorem 63A. Any quotient space is homeomorphic to partition.

Any quotient space Y induced by $q : X \twoheadrightarrow Y$ is homeomorphic to some X/E.
Proof.

let $E \subseteq X^2$ be the *equivalence kernel* of $q : X \twoheadrightarrow Y$

$$xEy \iff q(x) = q(y)$$

let $f([x]) = q(x)$ be well-defined injection (by def of E)
$\therefore f : X/E \xrightarrow{\sim} Y$ and $f \circ \pi = q$ (because q surjective)

$$\begin{tikzcd} X \arrow[r, two heads, "q"] \arrow[d, two heads, "\pi"'] & Y \\ X/E \arrow[ur, tail, dashed, "f"] & \end{tikzcd}$$

$\because q, \pi$ are quotient maps
$\therefore U \subseteq X/E$ open $\iff f[U] = q[\pi^{-1}[U]]$ open $\square$

Theorem 63B. When restriction of quotient map is quotient.

For quotient map $p : X \to Y$ and p-saturated subspace $A \subseteq X$, if A or p is open/closed, then $q = p \restriction A : A \twoheadrightarrow p[A]$ is quotient map.
Proof.

We first show

$$\forall V \subseteq p[A] \quad q^{-1}[V] = p^{-1}[V]$$
$$\forall U \subseteq X \quad p[U \cap A] = p[U] \cap p[A]$$

- To show $q^{-1}[V] = p^{-1}[V]$, notice $p^{-1}[V] \subseteq p^{-1}[p[A]] = A$.
- $p[U \cap A] \subseteq p[U] \cap p[A]$ follows by definition. For reverse inclusion, let any $p(u) = p(a) \in p[U \cap A]$. Then $u \in p^{-1}(p(a)) \subseteq A$, so $u \in U \cap A$.

Notice q is continuous, so it is left to show

$$\forall V \subseteq p[A] \quad q^{-1}[V] = p^{-1}[V] \text{ open} \in A \implies V \text{ open in } p[A].$$

Assume $A \subseteq X$ is open.
$\therefore p^{-1}[V]$ is open in X, so V is open in Y. ($p : X \to Y$ quotient map)
$\therefore V = V \cap p[A]$ open in $p[A] \subseteq Y$ as wanted.

Assume $p : X \to Y$ is open.
$\because p^{-1}[V] = U \cap A$ for some U open in X.
$\therefore V = p[p^{-1}[V]] = p[U \cap A] = p[U] \cap p[A]$ is open in $p[A]$. (p surjective)

□

7 Connected and Path Connected Spaces

7.1 Connectedness

Definition 71A. Connected space

for space X, a **separation** is (U, V) such that

$$X = U \sqcup V \text{ and } U, V \text{ open in } X$$

space X is **connected** iff has no separation

Lemma 71A. Space X is connected iff clopen sets are trivial.

Proof. $\varnothing \subset A \subset X$ clopen $\iff (A, X - A)$ separates □

Lemma 71B. Subspace Y of X is connected iff $\exists A \sqcup B = Y$ such that

$$A' \cap B = \varnothing = A \cap B'.$$

Proof. ($\Longleftarrow$)

Assume Y is separable by (A, B)
$\therefore A = \operatorname{cl}_X A \cap Y$, thus $A' \cap B \subseteq \operatorname{cl}_X A \cap B = \varnothing$ □

Proof. ($\Longrightarrow$)

Assume $Y = A \sqcup B$ and $A' \cap B = \varnothing = A \cap B'$.
$\therefore \operatorname{cl}_X A \cap B = \varnothing = A \cap \operatorname{cl}_X B$.

$\therefore \operatorname{cl}_X A \cap Y = A$ and $\operatorname{cl}_X B \cap Y = B$, thus both sets are closed.
$\therefore A = Y - B$ and $B = Y - A$ are open in Y. □

Lemma 71C. Connect subspace of disconnected space must be in one.

If (A, B) separates X and subspace $Y \subseteq X$ is connected, then $Y \subseteq A$ or $Y \subseteq B$.
Proof.

$\because A \cap Y$ and $B \cap Y$ are open in Y.
$\therefore$ both empty implies $(A \cap Y, B \cap Y)$ separates Y.
$\therefore A \cap Y$ or $B \cap Y$ is empty □

Theorem 71D. Union of connected with common point is connected.

For connected subspaces $\{A_\alpha\}$ of X,

$$\exists a \in \bigcap_\alpha A_\alpha \implies \bigcup_\alpha A_\alpha \text{ is connected.}$$

Proof.

BWOC assume (U, V) separates $A = \bigcup_\alpha A_\alpha$.
WLOG assume $a \in U$.
$\therefore a \in A_\alpha \subseteq U$ for any α. (Lemma 71C)
$\therefore A \subseteq U$, contradicts V nonempty. □

Theorem 71E. Closure of connected is connected

For connected space A,

$$A \subseteq B \subseteq \overline{A} \implies B \text{ is connected}$$

Proof.

BWOC assume (U, V) separates B, WLOG assume $A \subseteq U$. (A connected)
$\therefore \overline{A} \subseteq \overline{U}$, where $\overline{U} = U$ disjoints V.
$\therefore B \subseteq \overline{A} \subseteq U$ disjoints V, contradicts V nonempty. □

Theorem 71F. Image of connected is connected.

For continuous $f : X \to Y$,

$$A \subseteq X \text{ connected} \implies f[A] \subseteq Y \text{ connected.}$$

Proof.

Let $g : A \twoheadrightarrow f[A]$ be restriction of f (continuous).
BWOC assume (U, V) separates $g[A] = f[A]$.
$\therefore g^{-1}[U]$ and $g^{-1}[V]$ are
- disjoint, open in A (g is function, continuous)
- nonempty, union is A (g is surjective)

$\therefore$ contradicts that A is connected $\square$

Theorem 71G. If spaces X and Y are connected, then $X \times Y$ is connected.

Proof.

Fix $(a, b) \in X \times Y$, we reconstruct $X \times Y$ as double union of connected spaces.

Notice $X \times \{b\} \cong X$ and $\{x\} \times Y \cong Y$ are connected (for any $x \in X$).
Define $T_x = X \times \{b\} \cup \{x\} \times Y$ be connected (x, b) is common point
$\therefore X \times Y = \bigcup_{x \in X} T_x$ is connected (a, b) is common point

$\square$

Theorem 71H. Product of connected is connected

For connected spaces $\{X_\alpha : \alpha \in \Lambda\}$,

$$X = \prod_{\alpha \in \Lambda} X_\alpha \text{ connected.}$$

Proof.

Fix $a \in X$, we consturct a connected subspace $Z \subseteq X$ and show that $\overline{Z} = X$.

For $F \in [\Lambda]^{<\omega}$, define

$$X_F = \{x \in X : \forall(\alpha \in \Lambda)(x_\alpha \neq a_\alpha \implies \alpha \in F)\}.$$

In other words, $\forall \alpha \in \Lambda - F \quad x_\alpha = a_\alpha$.
$\therefore X_F \cong \prod_{\alpha \in F} X_\alpha$ is connected (finite product)
Define $Z = \bigcup_{F \in [\Lambda]^{<\omega}} X_F$ be connected (a is common point)

We show $X \subseteq \overline{Z}$, let any $x \in X$, any $U = \prod_\alpha U_\alpha \in \eta(x)$.
$\therefore \exists F \in [\Lambda]^{<\omega} \quad \forall \alpha \in F \quad U_\alpha = X$.
$\therefore$ let $z_\alpha = x_\alpha$ for any $\alpha \in F$, otherwise $z_\alpha = a_\alpha$, we have that $z \in U \cap X_F$. $\square$

Example 71A. $\mathbb{R}^\omega_\square$ is not connected.

Proof.

Let $A = \ell^\infty$ be set of bounded sequences, let B be unbounded sequences.
We show A, B are open.
For any $x \in \mathbb{R}^\omega$, let

$$U = \prod_i B(x, 1) \in \eta(x).$$

$\therefore$ if $x \in A$, then $U \subseteq A$; if $x \in B$, then $U \subseteq B$.
$\therefore$ (A, B) separates $\mathbb{R}^\omega_\square$ $\square$

7.2 Path Connectedness

Definition 72A. Path connected

For space X, a **path** from $a \in X$ to $b \in X$ is a continuous map $f : [0, 1] \to X$ such that

$$f(0) = a \quad f(1) = b.$$

- Tuple $(a, b) \in X^2$ is **path-connected** iff there exists such path.
- Space X is **path-connected** iff every tuple is path connected.

Notation. $f : a \rightsquigarrow b$ informally denotes a path from $a \in X$ to $b \in X$.

Corollary 72A. If X is path-connected, then X is connected.

Proof.

Let $X = A \sqcup B$ be separation.
Let any path $f : [0, 1] \to X$, thus $\operatorname{ran} f \subseteq X$ is connected space.
WLOG assume $\operatorname{ran} f \subseteq A$, thus f can't connect to any point in B. $\square$

Example 72A. Topologist's sine curve: connected but not path connected.

Let $S = \{(x, \sin(1/x)) : 0 < x < 1\}$.
$\because$ $f(x) = (x, \sin(1/x))$ is compont-wise continuous, thus continous.
$\therefore$ $S \subseteq \mathbb{R}^2$ is connected.
However, $\overline{S} = S \cup \{0\} \times [-1, 1]$ is connected but not path connected.

Theorem 72B. If $\{X_\alpha\}$ are path connected, then $X = \prod_\alpha X_\alpha$ is path connected.

Proof.

Let $x, y \in X$, thus $\forall \alpha \exists f_\alpha : x_\alpha \rightsquigarrow y_\alpha$.
Define $f : [0, 1] \to X$ by $f(t) = (f_\alpha(t))$, thus f continuous (f_α continuous)
$\therefore$ $f : x \rightsquigarrow y$. $\square$

7.3 Components, Path Components

Definition 73A. Compontents, Path components

For $x, y \in X$, define

$$x \underset{c}{\sim} y \iff \exists A \subseteq X \text{ connected} \quad x, y \in A$$
$$x \underset{pc}{\sim} y \iff \exists f : x \rightsquigarrow y$$

$X/\underset{c}{\sim}$ are **components** of X.
$X/\underset{pc}{\sim}$ are **path components** of X.

Corollary 73A. Each component is closed.

since closure of connected is connected

- $X/\underset{c}{\sim}$ is finite, then each component is also open.

Theorem 73B. Each nonempty (path) connected space only intersects one (path) component.

Proof. If (path) connected A intersects (path) connected C_1 at x_1 and C_2 at x_2, then $x_1 \sim x_2$, thus $C_1 = C_2$. □

Theorem 73C. Each (path) component is (path) connected.

Proof. We show $C \in X/\underset{c}{\sim}$ is connected.

Fix $a \in C$, then for any $x \in C$, choose A_x connected and has a, x.
$\therefore \forall x \in X \quad A_x \subseteq C$, thus $C = \bigcup_x A_x$ is connected (a is common point) □

7.4 Locally Connected

Definition 74A. A **local basis** at $x \in X$ is $\mathcal{B} \subseteq \eta(x)$ such that

$$\forall U \in \eta(x) \quad \exists B \in \mathcal{B} \quad B \subseteq U$$

Space X is **(path) locally connected** iff every point has a local basis consisting (path) connected sets.

Theorem 74A. Space X is locally (path) connected iff every (path) component of every open set is open.

Proof.

($\Longrightarrow$) For any $x \in C$, choose V as per definition, thus $V \subseteq C$.

($\Longleftarrow$) Let any $U \in \eta(x)$, let $C \subseteq U$ be the component of x in U.
$\therefore C$ is open in X, thus $C \in \eta(x)$ □

Theorem 74B. Space X is locally path connected iff each path component of U is open in X.

Proof. TODO □

Theorem 74C. If a space is locally path connected, then each component is a path component, and vice versa.

Proof. Let any component C of X, any $x \in C$.
Let P be the path component of x.
$\because$ P connected, thus $P \subseteq C$.
We now show $P = C$. BWOC assume $P \subset C$.
Let $\mathcal{Q} = \{P' \in X/\sim_{pc} : P' \neq P \quad P' \cap C \neq \varnothing\}$ and $Q = \bigcup \mathcal{Q}$.
- Notice $P' \subseteq C$ for any $P' \in \mathcal{Q}$.

$\therefore$ $C = P \sqcup Q$.
$\because$ P, Q are open. (X is locally path connected)
$\therefore$ P, Q disconnects C, contradicts C is connected. □

8 Filters, Ultrafilters

8.1 Filters, Ultrafilters

Definition 81A. Collection $\mathcal{A} \subseteq \mathcal{P}(X)$ has the **finite intersection property** (FIP) iff every nonempty finite subcollection has nonempty intersection.

$$\forall \mathcal{A}_0 \in [\mathcal{A}]^{<\omega} \quad \bigcap \mathcal{A}_0 \neq \varnothing.$$

$\mathcal{A}$ has the **strong finite intersection property** (SFIP) iff its infinite.

$$\forall \mathcal{A}_0 \in [\mathcal{A}]^{<\omega} \quad \left|\bigcap \mathcal{A}_0\right| \geqslant \omega.$$

FIP will come up in compactness later.

Definition 81B. Collection $\mathcal{F} \subseteq \mathcal{P}(X)$ is a **filter** on X iff
1. $X \in \mathcal{F}$ and $\varnothing \notin \mathcal{F}$.
2. $A, B \in \mathcal{F} \Longrightarrow A \cap B \in \mathcal{F}$.
3. $A \in \mathcal{F}, B \subseteq A \Longrightarrow B \in \mathcal{F}$.

Filter $\mathcal{U} \subseteq \mathcal{P}(X)$ is an **ultrafilter** on X iff
4. $\forall A \in \mathcal{P}(X) \quad A \in \mathcal{U} \Longleftrightarrow X - A \notin \mathcal{U}$.

Remark 81A. Every filter has FIP.

Remark 81B. Filter is ultrafilter iff is $\subseteq$-maximal.

Definition 81C. The **principal ultrafilter** on $x \subseteq X$ is

$$\mathcal{U}_x = \{A \subseteq X : x \in A\}$$

- ultrafilter is **free** iff not principal

Definition 81D. If $\mathcal{A}$ has FIP, then the filter **generated** by $\mathcal{A}$ is $\mathcal{F}_{\mathcal{A}} \supseteq \mathcal{A}$ where

$$\mathcal{F}_{\mathcal{A}} = \left\{A \subseteq X : \exists \mathcal{A}_0 \in [A]^{<\omega} \quad \bigcap \mathcal{A}_0 \subseteq A\right\}$$

Definition 81E. The **Fréchet filter** on X is

$$\mathsf{Fr} \stackrel{\text{def}}{=} \{A \subseteq X : A \text{ cofinite}\}$$

Theorem 81A. If $\mathcal{U}$ is an ultrafilter on X, then TFAE

- $\mathcal{U}$ is principal (i.e. not free).
- Some element of $\mathcal{U}$ is a finite set.
- Some element of $\mathcal{U}$ is a singleton.
- $\mathcal{U}$ is principal at some point. i.e. $\bigcap \mathcal{U} = \{x\}$
- $\mathsf{Fr} \not\subseteq \mathcal{U}$

Lemma 81B. If $\mathcal{C}$ is a chain of filters, then $\bigcup \mathcal{C}$ is a filter.

Theorem 81C. (AC) every filter $\mathcal{F} \subseteq \mathcal{P}(X)$ can be extended to an ultrafilter.

Proof. Consider the poset $(P, \subseteq)$

$$P = \{G \subseteq \mathcal{P}(X) : G \supseteq F \text{ is filter}\}$$

$\because$ every chain $C \subseteq P$ is bounded by $\bigcup C$.
$\therefore$ P has a maximal element (Zorn's), which is the desired ultrafilter. $\square$

8.2 Stone-Čech Compactification

Definition 82A. Let

$$\beta\omega \stackrel{\text{def}}{=} \{U \subseteq \mathcal{P}(\omega) : U \text{ is ultrafilter}\}$$

endowed with topology generated by

$$[A] = \{U \in \mathcal{P}(\omega) : A \in U\}$$

8.3 Convergence on Filters

Definition 83A. For space X and filter $\mathcal{F} \subseteq \mathcal{P}(\mathbb{N})$, a sequence $(x_n \in X)$ is said to $\mathcal{F}$ **converge** to $x \in X$ (denoted by $x_n \to_{\mathcal{F}} x$) iff

$$\forall U \in \eta(x) \quad \{n : x_n \in U\} \in \mathcal{F}.$$

The **limit points** of (x_n) under $\mathcal{F}$ is

$$\lim_{\mathcal{F}} x_n \stackrel{\text{def}}{=} \{x \in X : x_n \to_{\mathcal{F}} x\}$$

Lemma 83A. $x_n \to x$ iff $x_n \to_{\mathsf{Fr}} x$

Corollary 83B. If $x_n \to x$, then for any free ultrafilter $\mathcal{U}$, we have $x_n \to_{\mathcal{U}} x$.

$$\forall \mathcal{C} \subseteq \mathcal{T} \left(\bigcup \mathcal{C} = X \implies \exists \mathcal{C}_0 \in [\mathcal{C}]^{<\omega} \left(\bigcup \mathcal{C}_0 = X \right) \right)$$

9 Tychnoff's Theorem, Compactification

Theorem 9A. (Alexander) If space X has subbasis $\mathcal{S}$ where each open cover $\mathcal{C} \subseteq \mathcal{S}$ has a subcover, then X is compact.

Proof. BWOC assume X has open cover $\mathcal{C} \subseteq \mathcal{P}(X)$ with no finite subcover.
WLOG assume X is $\subseteq$ maximal (exists by Zorn's). We contradict by constructing a finte subcover.
$\because$ $\mathcal{C} \cap \mathcal{S}$ cannot cover X.
$\therefore$ $\exists x \in X - \bigcup \mathcal{C} \cap \mathcal{S}$ and $\exists U \in \eta(x) \cap \mathcal{C}$, and for some subbasic $S_n \in \mathcal{S}$

$$\bigcap_n S_n \subseteq U.$$

WLOG assume $U \in \mathcal{S}$ and $S_i \notin \mathcal{C}$ (otherwise $\mathcal{C} \cap \mathcal{S}$ covers X by AC).
For each i, let $\mathcal{C}_i = \mathcal{C} \cup \{S_i\}$ and $\mathcal{F}_i \cup \{S_i\} \in [\mathcal{C}_i]^{<\omega}$ be finite subcover ($\mathcal{C}$ was chosen to be maximal).
$\therefore$ $\bigcup \mathcal{F}_i \cup \{\bigcap S_i\}$ finitely covers X.
$\because$ $U \supseteq \bigcap S_i$ and $U \in \mathcal{C}$.
$\therefore$ $\mathcal{F} = \bigcup \mathcal{F}_i \cup \{U\} \in [\mathcal{C}]^{<\omega}$ covers X. □

Theorem 9B. (Tychnoff) If each X_α compact, then $X = \prod_\alpha X_\alpha$ compact.

Proof. We apply Alexander's with the subbasis

$$\mathcal{S} = \{\pi_\alpha^{-1}[U] : U \text{ open in } X_\alpha\}.$$

Let any open cover $\mathcal{C} \subseteq \mathcal{S}$.
We show $\exists \alpha$ such that $\mathcal{C}_\alpha = \{\pi_\alpha^{-1}[U] \in \mathcal{C} : U \subseteq X_\alpha\}$ covers X

- BWOC assume $\forall \alpha \exists x_\alpha \in X$ such that

$$\forall \pi_\alpha^{-1}[U] \in \mathcal{C}_\alpha \quad x_\alpha \notin U$$

$\therefore$ (x_α) cannot covered by $\mathcal{C}$, contradicts $\mathcal{C}$ covers.

Let $\mathcal{D} = \{U \subseteq X_\alpha : \pi_\alpha^{-1}[U] \in \mathcal{C}_\alpha\}$ covers X_α, let $\mathcal{D}_0 \in [\mathcal{D}]^{<\omega}$ be subcover.
$\therefore$ $\{\pi_\alpha^{-1}[U] : U \in \mathcal{D}_0\} \in [\mathcal{C}]^{<\omega}$ covers X. □

Theorem 9C. Tychnoff $\implies$ AC

Proof.

Let nonempty sets X_α, we show $X = \prod_\alpha X_\alpha$ nonempty.
Let $y = \bigcup_\alpha X_\alpha \notin \bigcup_\alpha X_\alpha$.
Consider topology on $Y_\alpha = X_\alpha \cup \{y\}$ defined as cofinite and $\{y\}$ open.
$\therefore$ each Y_α compact (only finitely many elements cannot be covered), thus $Y = \prod_\alpha Y_\alpha$ compact.
We show $\{\pi_\alpha^{-1}[X_\alpha]\}$ is FIP closed in Y.

1. X_α is closed in Y_α, so $\pi^{-1}[X_\alpha]$ closed in Y_α
2. For finite $\pi_n^{-1}[X_n]$, choose $x_n \in X_n$ (finite choce), thus
$$(y, \cdots, y, x_1, y, \cdots, y, x_2, y, \cdots, y, x_n, y, \cdots, y) \in \bigcap \pi_n^{-1}[X_n]$$

$\therefore \exists x \in \bigcap_\alpha \pi_\alpha^{-1}[X_\alpha]$, thus $x \in X$. □

10 Countability Axioms

Definition 10A. (First countable) Space X has a **countable basis** at $x \in X$ iff there is $\mathcal{B} = \{B_n \in \eta(x)\}$ such that
$$\forall U \in \eta(x) \quad \exists B_n \in \mathcal{B} \quad B_n \subseteq U.$$
A space is **first countable** iff every point has a countable basis

Remark 10A. Every metrisable space is first countable.

Theorem 10A. For space X

1. If some sequence in A converges to $x \in X$, then $x \in \overline{A}$
2. If $f : X \to Y$ is continuous, then for every convergent $x_n \to x$, the sequence $f(x_n) \to f(x)$.

The converse holds if X is metrisable

Definition 10B. (Second countable) A space X is **second countable** iff X has a countable basis.

Lemma 10B. 2ctb $\implies$ ccc $\implies$ discrete subspace ctb

Lemma 10C. If X is second countable, then every discrete subspace A of X is countable.

Example 10A. $\mathbb{R}^\omega_{\text{unif}}$ is metrisable (thus first countable) but not second countable.

11 Separation Axioms

12 Urysohn, Stock-Čech

13 Problem Set I

Example 13A. (4) intersection is topology

if $\{\mathcal{T}_\alpha\}_{\alpha\in\Lambda}$ is topology, then $\mathcal{T} = \bigcap_\alpha \mathcal{T}_\alpha$ is topology

Proof.

1. $\forall\alpha \quad \varnothing, X \in \mathcal{T}_\alpha$, so $\varnothing, X \in \mathcal{T}$
2. for any $\mathcal{S} \subseteq \mathcal{T}$
   - let any $\alpha \in \Lambda$
     - $\therefore \mathcal{S}$ is family of sets open in $\mathcal{T}_\alpha$
     - $\therefore \bigcup \mathcal{S} \in \mathcal{T}_\alpha$
   - $\therefore \bigcup \mathcal{S} \subseteq \bigcap_\alpha \mathcal{T}_\alpha$
3. for any $U_1, U_2 \in \mathcal{T}$, similiarly $U_1 \cap U_2 \in \mathcal{T}$ □

Example 13B. (6) π-basis

for space $(X, \mathcal{T})$

- $\mathcal{P} \subseteq \mathcal{T}^*$ is a **π-basis** iff

$$\forall U \in \mathcal{T}^* \quad \exists P \in \mathcal{P} \quad P \subseteq U$$

Proof. $\mathcal{P} = \{[p, q) : p, q \in \mathbb{Q}\}$ is **π-basis** but not basis of $\mathbb{R}_l$

- is π-basis because density

BWOC assume $\left[\sqrt{2}, 2\right) = \bigcup_\alpha [p_\alpha, q_\alpha)$

$\therefore \exists\alpha \quad \sqrt{2} \in [p_\alpha, q_\alpha)$

$\therefore p_\alpha < \sqrt{2}$ $\qquad$ ($p_\alpha \leqslant \sqrt{2}$ but $p_\alpha \in \mathbb{Q}$)

$\therefore \exists q \in \mathbb{Q} \quad p_\alpha < q < \sqrt{2}$

$\therefore q \notin \left[\sqrt{2}, 2\right)$, but $q \in [p_\alpha, q_\alpha) \subseteq \bigcup_\alpha [p_\alpha, q_\alpha)$ □

Example 13C. (9) topology generated by $\mathcal{B}$ is smallset topology containing $\mathcal{B}$

Proof.

let $\mathcal{T}_\mathcal{B}$ be topology generated by $\mathcal{B}$

let any topology $\mathcal{T} \supseteq \mathcal{B}$

$\therefore$ basis elements are open in $\mathcal{T}$

$\therefore$ unions of basis elements are open in $\mathcal{T}$

$\therefore \mathcal{T}_\mathcal{B} \subseteq \mathcal{T}$ □

Example 13D. (10) fursternberg topology

for $a \in \mathbb{Z}^*$ and $b \in \mathbb{Z}$, let $S(a, b) \stackrel{\text{def}}{=} \{ax + b : x \in \mathbb{Z}\}$

Proof. $\{S(a, b)\}$ is basis on $\mathbb{Z}$

1. $\forall n \in \mathbb{Z} \quad n \in S(0, n)$

2. for any $x_0 \in S(a_1, b_1) \cap S(a_2, b_2)$

$$\therefore \exists x_1 \exists x_2 \quad x_0 = a_1 x_1 + b_1 = a_2 x_2 + b_2$$
$$\therefore x_0 \in S(a_1 a_2, x_0) \subseteq S(a_1, b_1) \cap S(a_2, b_2)$$

□

Proof. $|S(a, b)| = \aleph_0$

$x \mapsto ax + b : \mathbb{N} \hookrightarrow S(a, b)$ since $a \neq 0$ □

Proof. $\mathbb{Z} - S(a, b)$ is open

$$\because k \in S(a, b) \iff k \equiv b \bmod a$$
$$\therefore \mathbb{Z} - S(a, b) = \bigcup_{\substack{n \in \mathbb{N} \\ n \not\equiv b \bmod a}} S(a, b)$$

□

Proof. $\bigcap_{p \text{ prime}} \mathbb{Z} - S(p, 0) = \{-1, 1\}$

wts for any $n \in \mathbb{Z}$

$$\forall p (n \notin S(p, 0)) \iff n \in \{-1, 1\}$$

($\Longrightarrow$) assume $n \notin \{-1, 1\}$, thus $|n| \neq 1$

- case $|n| < 1$, thus $|n| = 0$, so $n = 0 \in S(2, 0)$
- case $|n| > 1$, thus exist prime powers $q_1, \cdots, q_k$ such that (FToA)
$$|n| = q_1 \cdots q_k$$
  - where $q_i = p_i^{n_i}$ for some prime p_i and $n_i \in \omega$
- $\therefore n \in S(p_1, 0)$

($\Longleftarrow$) let any prime p

- $\therefore p > 1$ and we have $1, -1 \notin S(p, 0) = \{0, p, 2p, 3p, \cdots\}$
- $\therefore 1, -1 \in \bigcap_{p \text{ prime}} \mathbb{Z} - S(p, 0)$

□

Proof. there are infinitely many primes

BWOC assume only finitely many primes

$\therefore \{-1, 1\} = \bigcap_{p \text{ prime}} \mathbb{Z} - S(p, 0)$ is finite intersection of open, thus open

$\therefore \{-1, 1\}$ is union of some basis elements, contradicts $|S(a, b)| = \aleph_0$ □

14 Problem Set II

Example 14A. (9) $(\omega_1, \leqslant)$ is not discrete space

let woset $(X, \leqslant)$

Proof. (a) every nonmaximal element of X has an immediate successor

let any nonmaximal $x \in X$
$\therefore A = \{a \in X : a > x\}$ is nonempty
$\therefore \min A$ is immediate successor of x □

Proof. (b) every bounded subset of X has a supremum

let any subset $A \subseteq X$ bounded by $b \in X$
$\therefore B = \{b' \in X : b \text{ is a upper bound of } A\}$ is nonempty
$\therefore \min B = \sup A$ □

Proof. (c) ω_1 has no largest element

BWOC assume $\alpha = \max \omega_1$
$\therefore S_\alpha = \omega_1 - \{\alpha\}$ is countable, contradicts that S_α is countable □

Proof. (d) $A = \{\alpha \in \omega_1 : \alpha \text{ has no immediate predecessor}\}$ is uncountable

BWOC assume A countable
$\therefore S = \bigcup_{\alpha \in A} S_\alpha$ is countable (countable union of countable)
$\therefore \exists \beta_0 \in \omega_1 - S$ (ω_1 countable)
wts β_0 is upper bound of A
- BWOC assume $\exists \alpha > \beta_0$, then $\beta_0 \in S_\alpha \subseteq S$, contradicts $\beta_0 \notin S$

for any $n \in \omega$, let $\beta_{n+1} \in \omega_1$ be immediate successor of β_n (ω_1 has no max)
$\therefore B = \{\beta_n\}_{n \in \omega} \subseteq \omega_1$ is countable
$\therefore$ similarly using $\bigcup_n S_{\beta_n}$, it is bounded, thus has supremum γ

wts $\gamma \notin A$ has has no immediate predeecssor
- BWOC assume $\delta < \gamma$ is imm pred
  $\therefore \exists \beta_n \in B \quad \delta < \beta_n \leqslant \gamma$ ($\gamma = \sup B$)
  $\therefore \beta_n = \gamma < \beta_{n+1}$, contradicts γ is upper bound (δ is *imm* pred)

□

Proof. (e) $(\omega_1, <)$ is not discrete space

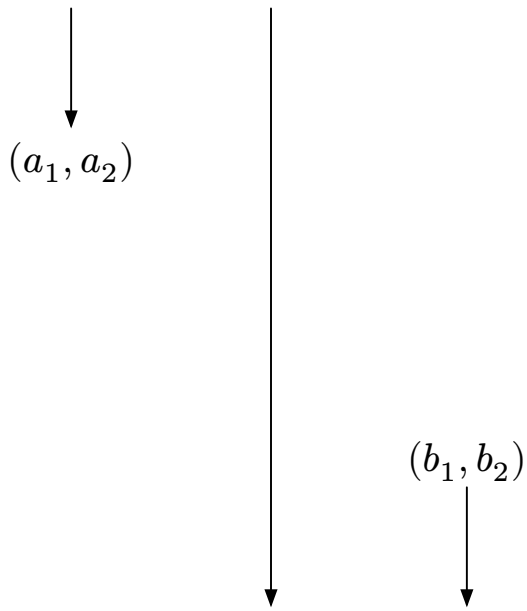
BWOC assume discrete
$\therefore$ pick some $\alpha \in \omega_1$ with no immediate predecessor, then $\{\alpha\}$ is open but $\{\alpha\} \neq (a, b)$ for any $a, b \in \omega_1$ □

15 Problem Set III

Example 15A. (1) $\mathbb{R}^2_{\text{lex}}$ is same topology as $\mathbb{R}_d \times \mathbb{R}$

Proof. $\mathbb{R}^2_{\text{lex}} \subseteq \mathbb{R}_d \times \mathbb{R}$

let any $x \in \mathbb{R}^2$, any basic $B_1 = (a, b) \in \eta_{\text{lex}}(x)$
$\therefore a_1 \leqslant x_1 \leqslant b_1$, thus we have three cases:



case $a_1 = x < a_2$, let $B_2 = \{x\} \times (a_2, \infty)$
case $a_1 < x < a_2$, let $B_2 = \{x\} \times \mathbb{R}$
case $a_1 < x = a_2$, let $B_2 = \{x\} \times (-\infty, b_2)$
$\therefore x \in B_2 \subseteq B_1$ where B_2 is basic open in $\mathbb{R}_d \times \mathbb{R}$ □

Proof. $\mathbb{R}_d \times \mathbb{R} \subseteq \mathbb{R}^2_{\text{lex}}$

let any $x \in \mathbb{R}^2$, any basic open $B_2 = U \times (a_2, b_2) \in \eta_d(x)$
$\therefore x_1 \in U$ and $a_2 < x_2 < b_2$
$\therefore$ let $a_1 = b_1 = x_1$, we have that $x \in (a, b) \in \eta_{\text{lex}}(x)$ □

Example 15B. (2) otop on convex $A \subseteq (X, \leqslant)$ is same as subspace top of X

subset A of loset $(X, \leqslant)$ is **convex** iff

$$\forall a, b \in A \quad [a, b] \subseteq A$$

Proof.

let $\mathcal{T}_{\leqslant}$ be order topology of A, generated by usual basis $\mathcal{B}_{\leqslant}$
let $\mathcal{T}_A$ be subspace topology , generated by usual basis $\mathcal{B}_A$

wts $\mathcal{T}_{\leqslant}$ coarser than $\mathcal{T}_A$
- let any basic open $(a, b) \in \mathcal{T}_{\leqslant}$
- $\therefore [a, b] \subseteq A$, so $A \cap (a, b) \subseteq (a, b)$

wts $\mathcal{T}_A$ coarset than $\mathcal{T}_{\leqslant}$
- let any basic open $A \cap (a, b) \in \eta_A(x)$

- WLOG assume x is not max/min of A
- $\therefore \exists a_x, b_x \in A \quad a_x < x < b_x$
- $\therefore [a_x, x], [x, b_x] \subseteq A$, so $[a_x, b_x] \subseteq A$
- $\therefore \exists a', b' \in A \quad a \leqslant a' < x < b' \leqslant b$ (otherwise contradicts $[a_x, b_x] \subseteq A$)
- $\therefore x \in (a', b') \subseteq A \cap (a, b)$ □

Definition 15A. closed basis

family $\mathcal{C} \subseteq \mathcal{P}(X)$ is a **closed basis**

1. $\bigcap \mathcal{C} = \varnothing$
2. $\forall C_1, C_2 \in \mathcal{C} \quad \forall x \notin C_1 \cup C_2 \quad \exists C_3 \in \mathcal{C} \quad x \notin C_3 \supseteq C_1 \cup C_2$

Corollary. $\mathcal{C} \subseteq \mathcal{P}(X)$ is closed basis iff $\{X - C\}_{C \in \mathcal{C}}$ is (open) basis

16 Problem Set IV

Example 16A. (4?) continous in each variable, but not continuous

let $f : \mathbb{R}^2 \to \mathbb{R}$ be

$$f(0,0) = 0$$
$$f(x,y) = xy/(x^2 + y^2)$$

Proof. $f_x = \lambda y. f(x,y) : \mathbb{R} \to \mathbb{R}$ is continuous □

Proof. $f : \mathbb{R}^2 \to R$ is not continuous □

Example 16B. (6) subspace topology is initial topology of inclusion map

let space X, subset $A \subseteq X$
let subspace topology $\mathcal{T}_A$, initial topology $\mathcal{T}_i$
Proof. $\mathcal{T}_i \subseteq \mathcal{T}_A$

$\because \forall U \subseteq X \quad i^{-1}[U] = U \cap A \in \mathcal{T}_A$
$\therefore i : (A, \mathcal{T}_A) \to X$ continuous, so $\mathcal{T}_i \subseteq \mathcal{T}_A$ ($\mathcal{T}_i$ is by def coarest) □

Proof. $\mathcal{T}_A \subseteq \mathcal{T}_i$

$\because T_i$ is generated by $\mathcal{S} = \{i^{-1}[U] : U \in \mathcal{T}_X\}$ where $i^{-1}[U] = U \cap A$
$\therefore$ every basic $U \cap A \in \mathcal{T}_A$ is open in T_i, thus $\mathcal{T}_A \subseteq \mathcal{T}_i$ □

Example 16C. (7?) any topology is the init top of cont maps to Sierpinsky

let S be Sierpinsky
let any space $(X, \mathcal{T})$, let $\mathcal{F} = \{f \in S^X : f \text{ continuous}\}$
Proof. $\mathcal{T}$ is initial topology of $\mathcal{F}$ □

17 Problem Set V

Example 17A. (2)

Given metric space (X, d)

Proposition 17A. $d'(x, y) = d(x, y)/(1 + d(x, y))$ is metric.

Proof.

First, define $f : \mathbb{R} \to \mathbb{R}$ by

$$f(x) = \frac{x}{1+x} = 1 - \frac{1}{1+x}$$

Notice $f : \mathbb{R} \to [0, 1)$ is
- nonnegative, bounded, increasing,
- $f(x) = 0 \iff x = 0$.

Thus $f \circ d = d'$ is
- nonnegative, symmetric, bounded,
- zero $\iff d$ is zero $\iff x = y$.

To show triangle inequality, first note that for $a, b \geqslant 0$,

$$f(a+b) - f(b) = \frac{a}{(1+b)(1+a+b)} \leqslant \frac{a}{1+a} = f(a)$$

$$\therefore \qquad f(a+b) \leqslant f(a) + f(b)$$

Lastly, because f is increasing and $f \circ d = d'$

$$f(d(x,z)) \leqslant f(d(x,y) + d(y,z)) \leqslant f(d(x,y)) + f(d(y,z))$$

Therefore d' is metric □

Proposition 17B. d' generates the same topology as d

Proof.

$\because$ f is d-continuous and $f^{-1}(y) = 1/(1-y)$ is d'-continuous.
$\therefore$ $d' = f \circ d$ is d-continuous, and $d = f^{-1} \circ d'$ is d'-continuous. □

Example 17B. (3) (X, d) metrisable $\implies X^\omega$ metrisable

$$D(x, y) = \sup_{i \in \omega} \frac{\overline{d}(x_i, y_i)}{i}$$

Proposition 17C. Product is finer than metric.

Proof. Let any basic $B_D(x, \varepsilon)$ open in metric, we show $\exists V \subseteq B(x, \varepsilon)$ open in product.

Let $N \in \mathbb{N}$ such that $1/N < \varepsilon$.
Let V open in product defined as

$$V = B_d(x_1, \varepsilon) \times \cdots \times B_d(x_N, \varepsilon) \times \mathbb{R} \times \mathbb{R} \times \cdots$$

We show $V \subseteq B(x, \varepsilon)$, for any $y \in V$, notice $\overline{d}(x_i, y_i) \leqslant 1$

$$\because \forall i \geqslant N \quad \frac{\overline{d}(x_i, y_i)}{i} \leqslant \frac{1}{N}$$

$$\therefore D(x, y) \leqslant \max\left\{\overline{d}(x_1, y_1)/1, \cdots, \overline{d}(x_N, y_N)/N, 1/N\right\} < \varepsilon$$

□

Proposition 17D. Metric is finer than product.

Proof. Let any basic $U = \prod_{n \in \omega} U_n$ with $U_n = \mathbb{R}$ for any $n \notin \Lambda_0 \in [\omega]^{<\omega}$.
Let any $x \in U$, we show $\exists B(x, \varepsilon) \subseteq U$.
Choose $B(x_i, \varepsilon_i) \subseteq U_i$ for any $i \in \Lambda_0$ where $\varepsilon_i < 1$. Define

$$\varepsilon = \min\{\varepsilon_i / i\}$$

We show $B(x, \varepsilon) \subseteq U$, let any $y \in B(x, \varepsilon)$,

$$\therefore \forall i \in \Lambda_0 \quad \overline{d}(x_i, y_i) \leqslant iD(x, y) < i\varepsilon \leqslant \varepsilon_i < 1$$

Therefore $d(x_i, y_i) = \overline{d}(x_i, y_i) < \varepsilon_i$, so $y \in U$. □

Example 17C. (4)

(a) For function $f : X \to Y$

$$f \text{ continuous} \Longrightarrow (x_n \to x \Longrightarrow f(x_n) \to f(x))$$

Proof.

Let any $x_n \to x$. We show $f(x_n) \to f(x)$.
Let any $V \in \eta(f(x))$, thus $f^{-1}[V] \in \eta(x)$.
$\therefore \exists N \forall n > N \quad x_n \in f^{-1}[V]$, thus $f(x_n) \in V$ □

(b) converse holds for sequential spaces
Proof.

We show f continuous by $f[\overline{A}] \subseteq \overline{f[A]}$.

Let any $f(a) \in f[\overline{A}]$, any $V \in \eta(f(a))$.
$\therefore \exists (a_n) \subseteq A \quad a_n \to a.$ $(a \in \overline{A})$
$\therefore f(a_n) \to f(a)$, thus $\exists N \quad f(a_N) \in V \cap f[A]$. □

(c) counterexample of (b) when the space is not sequential

Let $\mathrm{id}_{\mathbb{R}} : \mathbb{R}_{\text{cocountable}} \to \mathbb{R}$
Proof.

We show in $\mathbb{R}_{\text{coco}}$, sequence $x_n \to x$ iff x_n is eventually x.
Assume x_n is not eventually x.
$\therefore \mathbb{R} - \{x_n : x_n \neq x\} \in \eta(x)$ but does not almost contain $\{x_n\}$. Thus $x_n \not\to x$.
Thus $x_n \to x \implies f(x_n) \to f(x)$, but f is not continuous becaue $f^{-1}[(0,1)]$ is not open □

Example 17D. (5) $\mathbb{R}_{\text{lex}}$ is metrisable.

Let $\overline{d} : \mathbb{R} \to \mathbb{R}$, define

$$D((x, y_1), (x, y_2)) = \overline{d}(y_1, y_2)$$
$$D((x_1, y_1), (x_2, y_2)) = 1 \qquad x_1 \neq x_2$$

Proof.

D is metric because $\overline{d}$ is metric.

We show D generates $\mathbb{R}_{\text{lex}}$. Consider only radius $\varepsilon < 1$.

1. $B_D((x, y), \varepsilon) = \{x\} \times B_{\overline{d}}(y, \varepsilon) = B_d(y, \varepsilon)$ open in $\mathbb{R}_{\text{lex}}$.
2. Every open set in $\mathbb{R}_{\text{lex}}$ is union of intervals on a certain x, which are balls. □

Example 17E. (7) Saturated Open $\implies$ Quotient Open

Let equivalence relation E on space X. Let $A \subseteq X$ and $R = E \cap A^2$.

Proposition 17E. (a) If A is E-saturated, then $A/R \subseteq X/E$.

Proof.

Let any $[a]_R \in A/R$. Obviously $[a]_R \subseteq [a]_E$.
For reverse inclusion, $aEb \implies b \in A$ because $A = \bigcup_a [a]_E$ □

Proposition 17F. (b) If A is E-saturated and open in X, then A/R is open in X/E.

Proof.

Let $\pi = x \mapsto [x]_E$ be projection, which is quotient map in X/E. Notice

$$\pi^{-1}[A/R] = \left\{b \in A : \exists a \in A \quad b \in [a]_R\right\}.$$

And $A \subseteq \pi^{-1}[A/R]$ because A is E-saturated. □

Example 17F. (8)

Let $\sim_1 \subseteq \sim_2$ on topology X. Let $X_1 = X/\sim_1$ and $X_2 = X/\sim_2$.
Let equivalence relation $\sim$ on X_1 defined by

$$\pi_1(x) \sim \pi_1(y) \text{ iff } x \sim_2 y$$

Let $Y = X_1/\sim$

Proposition 17G. $X_2 \cong Y$.

Proof. We show

$$\begin{tikzcd} X \arrow[r, two heads, "\pi_1"] \arrow[d, two heads, "\pi_2"'] & X_1 \arrow[d, two heads, "\pi"] \\ X_2 \arrow[r, tail, dashed, two heads, "f"] & Y \end{tikzcd}$$

Define $f(\pi_2(x)) = \pi(\pi_1(x))$, this is well-defined and injective because for any $a, b \in X$

$$\begin{aligned} \pi_2(a) = \pi_2(b) &\iff a \sim_2 b \\ &\iff \pi_1(a) \sim \pi_1(b) \\ &\iff \pi(\pi_1(a)) = \pi(\pi_1(b)) \end{aligned}$$

- $f : X_2 \to Y$ is surjective because $\pi \circ \pi_1$ is surjective.

We show f is continuous and open. For any $U \subseteq X_1$, since π_2, π_1, π are quotient maps,

$$\begin{aligned} U \text{ open} &\iff V = \pi_2^{-1}[U] \text{ open} \\ &\iff (\pi \circ \pi_1)[V] = f[U] \text{ open} \end{aligned}$$

□

Example 17G. (9) Sequential, Nonfirst-countable Space

Let $X = \mathbb{R}/\sim$ be rose with countably many pedals where

$$x \sim y \text{ iff } x = y \text{ or } x, y \in \mathbb{Z}$$

Proposition 17H. X is sequential.

Proof. Let any $A \subseteq X$, we show

$$\forall \pi(x) \in \overline{A} \quad \exists(\pi(x_n) \in A) \quad \pi(x_n) \to \pi(x). \tag{1}$$

Remark. To show Equation 1, it suffices to show that $x \in \overline{\pi^{-1}[A]}$ because then,

$\therefore \exists(x_n \in \pi^{-1}[A])$ such that $x_n \to x$. ($\mathbb{R}$ sequential)

$\therefore \pi(x_n) \to \pi(x)$ where $\pi(x_n) \in A$ (π continuous)

Case $\mathbb{Z} \in \overline{A}$, we show $\exists x \in \mathbb{Z} \cap \overline{\pi^{-1}[A]}$.
BWOC assume $K = \overline{\pi^{-1}[A]} \subseteq \mathbb{R} - \mathbb{Z}$.
$\therefore$ $\pi[K]$ is family of singletons, thus K is closed saturated.
$\therefore$ $\pi[K] \supseteq (\pi \circ \pi^{-1})[A] = A$ is closed. (π quotient map)
$\therefore$ $\mathbb{Z} \in \overline{A} \subseteq \pi[K]$, thus $\exists x \in \mathbb{Z} \cap K$, contradicts the assumption.
Thus $\mathbb{Z} = \pi(x)$ where $x \in \overline{\pi^{-1}[A]}$ as wanted.

Case $\pi(x) = \{x\} \in \overline{A}$ where $x \notin \mathbb{Z}$.
Let any basic $B(x, \varepsilon) \in \eta(x)$.
WLOG assume ε is small enough such that $B(x, \varepsilon) \subseteq \mathbb{R} - \mathbb{Z}$. ($\mathbb{R}$ Hausdorff)
$\therefore$ $B(x, \varepsilon)$ is open saturated, because the image is set of singletons.
$\therefore$ $\pi[B(x, \varepsilon)] \in \eta(\pi(x))$, thus it intersects A at some $\pi(y) = \{y\}$.
$\therefore$ $y \in B(x, \varepsilon) \cap \pi^{-1}[A]$.
Thus $x \in \overline{\pi^{-1}[A]}$ as wanted.

□

Proposition 17I. X is not first-countable.

Proof. (diagonalisation goes brrrr)

BWOC assume $\mathbb{Z} = \pi(0) \in X$ has a countable basis $\{U_n \in \eta(\mathbb{Z})\}$.
$\therefore$ $\forall n \in \mathbb{N} \quad \pi^{-1}[U_n] \in \eta(n)$ (π continuous)
For any $n \in \mathbb{N}$, choose $B(n, \varepsilon_n) \subseteq \pi^{-1}[U_n]$ and let $B_n = B(n, \varepsilon_n/2)$.
- WLOG assume $\{B(n, \varepsilon_n)\}$ is pairwise disjoint.

Define $U = \bigcup_{n \in \mathbb{Z}} B_n$ where $B_n = B(n, 1)$ for any $n \in \mathbb{Z} - \mathbb{N}$.
$\therefore$ $U \supseteq \mathbb{Z}$ is open saturated, thus $\pi[U] \in \eta(\mathbb{Z})$.
Thus $\exists n \in \mathbb{N}$ such that $\pi^{-1}[U_n] \subseteq U$.
However, $B(n, \varepsilon_n) \subseteq \pi^{-1}[U_n]$ whereas $B(n, \varepsilon_n) \nsubseteq U$ □

18 Problem Set VI

Example 18A. (1)

$$X = [0, 1]^2 \text{ order topology is connected but not path connected.}$$

Proof.

BWOC assume $f : [0, 1] \to X$ is a path from $(0, 0)$ to $(1, 1)$.

$\because$ missing any point in $X - \{(0, 0), (1, 1)\}$ will disconnect $(0, 0)$ and $(1, 1)$.
$\therefore$ $f : [0, 1] \twoheadrightarrow X$.
$\therefore$ for each $x \in [0, 1]$, the set

$$U_x = f^{-1}[\{x\} \times (0,1)]$$

is nonempty open in $[0,1]$.

Choose $q_x \in U_x \cap \mathbb{Q}$ for each $x \in [0,1]$.
$\because \{U_x : x \in [0,1]\}$ is pairwise disjoint.
$\therefore x \mapsto q_x$ is injective, contradicts $[0,1]$ is uncountable. $\square$

Example 18B. (2)

Let subspaces $(A_n \subseteq X)$ such that $A_n \cap A_{n+1} \neq \varnothing$ and

$$A := \bigcup_{n \in \mathbb{N}} A_n.$$

Proposition 18A. If A_n are connected, then A is connected.

Proof.

BWOC assume $A = U \sqcup V$ is separation, WLOG assume $A_1 \subseteq U$.
$\therefore A_n \subseteq U$ for any $n \in \mathbb{N}$, thus $A \subseteq U$ $\square$

Proposition 18B. If A_n are path connected, then A is path connected.

Proof.

Let any $x, y \in A$. We construct path $f : x \rightsquigarrow y$ in A.
WLOG assume $x \in A_n$ and $y \in A_m$ for some $n \leqslant m$.
Define $x_n = x$ and $x_{m+1} = y$. And for $n \leqslant k < m$, choose $x_{k+1} \in A_k \cap A_{k+1}$.
$\therefore x_k, x_{k+1} \in A_k$ for any $n \leqslant k \leqslant m$, choose path $f_k : [k, k+1] \to X$.

Define $f = f_n \cup \cdots \cup f_m$, thus $f : [n, m+1] \to A$ is continuous (pasting lemma) and $f : x \rightsquigarrow y$. $\square$

Example 18C. (3)

If $A \subseteq R^2$ is countable, then $\mathbb{R}^2 - A$ is path connected.

Proof. We proceed to construct a path $f : x \rightsquigarrow z' \rightsquigarrow y$.

For any $x, y \in \mathbb{R}^2$, define $L(x,y) \subseteq \mathbb{R}^2$ to be the line passing through x and y.
Define $L_x \subseteq \mathcal{P}(\mathbb{R}^2)$ to the the set of lines passing through x; notice L_x and $L(x,y)$ are uncountable.

Let $A_x = \{L(x,a) : a \in A\}$ be set of all lines through x that intersects A.
$\because A_x$ is countable, choose $L(x,z) \in L_x - A_x$.
Consider $L = \{L(z',y) : z' \in L(x,z)\}$, this set is uncountable.
$\therefore$ choose $L(z',y) \in L - A_y$, we have that $L(x,z'), L(z',y) \subseteq \mathbb{R}^2 - A$.

Let $f_x : x \rightsquigarrow z'$ and $f_y : z' \rightsquigarrow y$ be the paths induced by $L(x, z')$ and $L(z', y)$.
$\therefore f = f_x \cup f_y : x \rightsquigarrow y$ is a path in $\mathbb{R}^2 - A$ □

Example 18D. (5) Brouwer fixed-point theorem

If $f : [0, 1] \to [0, 1]$ is continuous, then f has a fixed point.
Proof.

Define continuous $g(t) = f(t) - t$ on domain $[0, 1]$.
$\therefore g(1) \leqslant 0 \leqslant g(0)$, thus $\exists t \in [0, 1]$ such that $f(t) - t = g(t) = 0$ (IVT)

□

Example 18E. (6) Borsuk–Ulam

Define S^1 to be unit circle

$$S^1 \stackrel{\text{def}}{=} \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$$

If $f : S^1 \to \mathbb{R}$ continuous, then $\exists s \in S^1$ such that $f(s) = f(-s)$.
Proof.

Define continuous $g(x) = f(x) - f(-x)$, notice $g(-x) = -g(x)$, that is, odd.
Choose an $x \in \mathbb{R}$, if $g(x) = 0$, we are done.
WLOG assume $g(x) > 0$, thus $g(-x) = -g(x) < 0$.
$\therefore \exists s \in [-x, x] \quad f(s) - f(-s) = g(s) = 0.$ □

Example 18F. (6)

Proposition 18C. $\mathbb{R} \ncong \mathbb{R}^2$

Proof. BWOC assume $f : \mathbb{R} \xrightarrow{\sim} \mathbb{R}^2$, then $\mathbb{R} - \{0\}$ is disconnected whereas its image is not. □

Proposition 18D. $(0, 1) \ncong [0, 1)$

Proof. $[0, 1) - \{0\}$ is connected whereas its image $(0, 1) - \{f(0)\}$ is not. □

Proposition 18E. $[0, 1) \ncong [0, 1]$

Proof. $[0, 1] - \{0, 1\}$ is connected whereas is image is not. (f injective) □

Proposition 18F. $[0, 1) \ncong S^1$

Proof. $[0, 1) - \{1/2\}$ is disconnected, but $S^1 - \{f(1/2)\}$ is not. □

Example 18G. (7) Totally disconnected

Space X is **totally disconnected** iff all components are singletons.

Proposition 18G. Sorgenfrey $\mathbb{R}_l$ is totally disconnected.

Proof. Let any $x, y \in S \subseteq \mathbb{R}_l$, WLOG assume $x < y$. Thus $(-\infty, y) \cap S, [y, \infty) \cap S$ disconnects S. $\square$

Remark 18A. All continuous maps $f : \mathbb{R} \to \mathbb{R}_l$ are constant maps, because $\mathbb{R}$ is connected (so does $f[\mathbb{R}]$)

Example 18H. (8)

Let $\mathbb{R}^\omega$ with uniform topology

Proposition 18H. The component of constant zero sequence is

$$\ell^\infty \stackrel{\text{def}}{=} \left\{ x \in \mathbb{R}^\omega : \sup_n |x_n| < \infty \right\}$$

$a^1 x^n a^i U^x$

$a_1 x_n a_i U_x$

Definition 18A. $\mathcal{A} \subseteq \mathbb{N}$ has **strong finite intersection property** (SFIP) iff every finite subset has an infinite intersection.

Example 18I. Q1

Definition 18B. For space X and filter $\mathcal{F}$ on $\mathbb{N}$. A sequnence (x_n) is said to $\mathcal{F}$-**converges** to $x \in X$ (denoted $x_n \to_{\mathcal{F}} x$) iff

$$\forall U \in \eta(x) \quad \{n \in \mathbb{N} : x_n \in U\} \in \mathcal{F}.$$

Definition 18C. For filter $\mathcal{F}$,

$$\lim_{\mathcal{F}} x_n \stackrel{\text{def}}{=} \{x \in X : x_n \to_{\mathcal{F}} x\}$$

Example 18J. Q2

(a) $x_n \to x$ iff $x_n \to_{\mathsf{Fr}} x$
Proof. Let any $U \in \eta(x)$

($\Longrightarrow$) $\therefore \exists N \forall n \geqslant N \quad x_n \in U$, thus $A = \{n \in \mathbb{N}, n \geqslant N\}$ cofinite, so $A \in \mathsf{Fr}$.
($\Longleftarrow$) $\therefore A = \{n \in \mathbb{N} : x_n \in U\}$ cofinite.
let $N = \max(\mathbb{N} - A)$ or 1, we have $\forall n \geqslant N \quad x_n \in U$. $\square$

(b) X compact, $\mathcal{U}$ ultrafilter, and $x_n \in X$, then $\lim_{\mathcal{U}} x_n$ nonempty.

Proof. BWOC assume $\lim_{\mathcal{U}} x_n = \varnothing$, we will contradict compactness.
$\therefore \forall x \in X \quad \exists U_x \in \eta(x) \quad \{n \in \mathbb{N} : x_n \in U_x\} \notin \mathcal{U}$, that is,
$\{n \in \mathbb{N} : x_n \notin U_x\} \in \mathcal{U}$.
Let $C = \{U_x : x \in X\}$ be open cover of X, let $C_0 \in [C]^{<\omega}$ be finite subcover.
$\because \{n \in \mathbb{N} : x_n \notin U_x\} \in \mathcal{U}$ for any $U_x \in C_0$.
Let $N \in \bigcap_{U_x \in C_0} \{n \in \mathbb{N} : x_n \notin U_x\} \in \mathcal{U}$.
$\therefore \forall U_x \in C_0 \quad x_N \notin U_x$, contradicts C_0 covers X. □

(c) X Hausdorff, $\mathcal{U}$ ultrafilter, $x_n \in X$, then $|\lim_{\mathcal{U}} x_n| \leqslant 1$.
Proof. Let any $x, y \in \lim_{\mathcal{U}} x_n$, separated by U_x, U_y.
$\therefore A = \{n : x_n \in U_x\}, B = \{n : x_n \in U_y\} \in \mathcal{U}$.
$\therefore \exists N \in A \cap B \in \mathcal{U}$, thus $x_N \in U_x \cap U_y$. So $x = y$.

□

Example 18K. Q3

$$X \subseteq \mathbb{R}_l \text{ compact} \Longrightarrow X \text{ countable}$$

Proof.

$\because \mathbb{R}$ coarser than $\mathbb{R}_l$.
$\therefore X$ also compact in $\mathbb{R}$, thus closed and bounded.

Show $(X, >)$ wosubset of $\mathbb{R}$ by
Let any strictly increasing sequence $x_n \in X$
$\therefore$ let $\alpha = \sup_n x_n, m = \min X \in X$ (X closed and bounded)
$\therefore [m, x_1), [x_n, x_{n+1}), [\alpha, \infty)$ covers X but has no finite subcover.
Show X countable by
$\forall x \in X$, let $s_x = \max[m, x) < x$. (woset)
$\therefore \{[s_x, x) : x \in X\}$ is pariwise disjoint collection of intervals of $\mathbb{R}$, thus countable.

□

Example 18L. Q4

Let X CHaus

(a) $\forall x \in X \quad \forall U \in \eta(x) \quad \exists V \in \eta(x) \quad \overline{V} \subseteq U$
Proof.

$\because X - U$ is closed, thus compact subspace
Let A, B open separates $x, X - U$.
Let $V = A \cap U \in \eta(x)$, BWOC assume $\exists y \in \overline{V} \cap X - U$.
$\therefore y \in \overline{V} \cap B$, so $B \in \eta(y)$ and intersects $V \subseteq A$, contradicts A, B disjoint □

(b) If $\{D_n\}$ are dense open, then $D = \bigcap_n D_n$ dense open

Proof. Choose an open $U \subseteq X$.
$\therefore \exists d_1 \in D_1 \cap U \quad \exists V_1 \in \eta(d_1) \quad \overline{V_2} \subseteq D_1 \cap U$.
$\therefore \exists d_2 \in D_2 \cap V_1 \quad \exists V_2 \in \eta(d_2) \quad \overline{V_2} \subseteq D_2 \cap V_1$.
$\therefore \exists d_{n+1} \in D_{n+1} \cap V_n \quad \exists V_{n+1} \in \eta(d_n) \quad \overline{V_{n+1}} \subseteq D_{n+1} \cap V_n$.
Notice $\overline{V_{n+1}} \subseteq V_n \subseteq \overline{V_n}$, so $\{\overline{V_n}\}$ is decreasing FIP family of closed sets.
$\therefore$ let $d \in \bigcap_n \overline{V_n}$, thus $d \in D \cap U$ □

Definition 18D. A space X is **zero-dimentional** iff it has a clopen basis.

Example 18M. Q5

For $n \in \mathbb{N}$ and $a \in 2^n$, define

$$[a] = \{x \in 2^{\mathbb{N}} : \forall i < n \quad x_i = a_i\}$$

Let $2 = \{0, 1\}$ under discrete topology

Proposition (a) Show $\{[a] : a \in 2^n \quad n \in \mathbb{N}\}$ is basis of $2^{\mathbb{N}}$
Proof.
- $[(0)], [(1)]$ covers $2^{\mathbb{N}}$
- If $[a]$ intersects $[b]$, WLOG a is prefix of b, thus $[b] \subseteq [a] \cap [b]$

□

Proposition (b) $2^{\mathbb{N}}$ is zero-dimentional.
Proof.

$$[a] = X - \bigcup_{\substack{b \in 2^{\mathbb{N}} \\ b \neq a}} [b]$$

thus $[a]$ clopen. □

Proposition (c) If X is zero-dimensional and T_1, then X is totally disconnected.
Proof. Let any distinct $x, y \in A$.
$\therefore X - \{x\} \in \eta_X(y)$, let clopen basic $B \subseteq X - \{x\}$ of y.
$\therefore A \cap B$ is a nontrivial clopen set of A, thus A disconnected. □

Proposition (d) $2^{\mathbb{N}}$ compact.
Proof. BWOC assume $\mathcal{C} \subseteq \mathcal{P}(2^{\mathbb{N}})$ has no finite subcover.
$\therefore \mathcal{C}$ can't fsubcov $[(x_1)]$ for some $x_1 \in 2$.
$\therefore \mathcal{C}$ can't fsubcov $[(x_1, x_2)]$ for some $x_2 \in 2$.
$\therefore \mathcal{C}$ can't fsubcov $[(x_1, \cdots, x_n)]$ for some $x_n \in 2$.
Let $U = \mathcal{C} \cap \eta(x)$, thus $\exists [(x_1, \cdots, x_n)] \subseteq U \quad [(x_1, \cdots, x_n)] \in \eta(x)$.
$\therefore$ contradicts that can't $\mathcal{C}$ fsubcov $[(x_1, \cdots, x_n)]$ □